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2018.7.2

航空宇宙設計最適化

Design optimization in aerospace engineering

大山　聖

Akira Oyama

ロケット開発における
多目的設計探査
MODE for rocket development

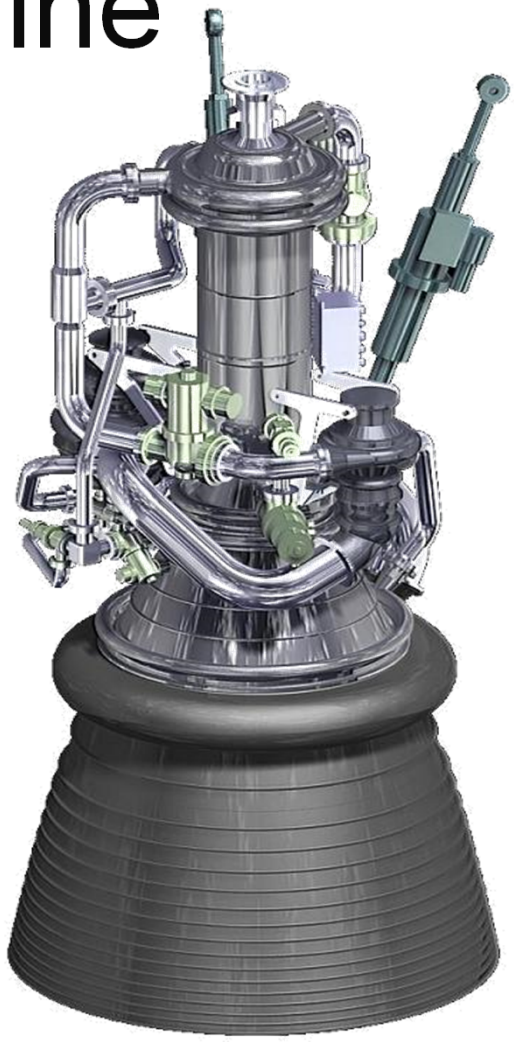
今日の内容 contents

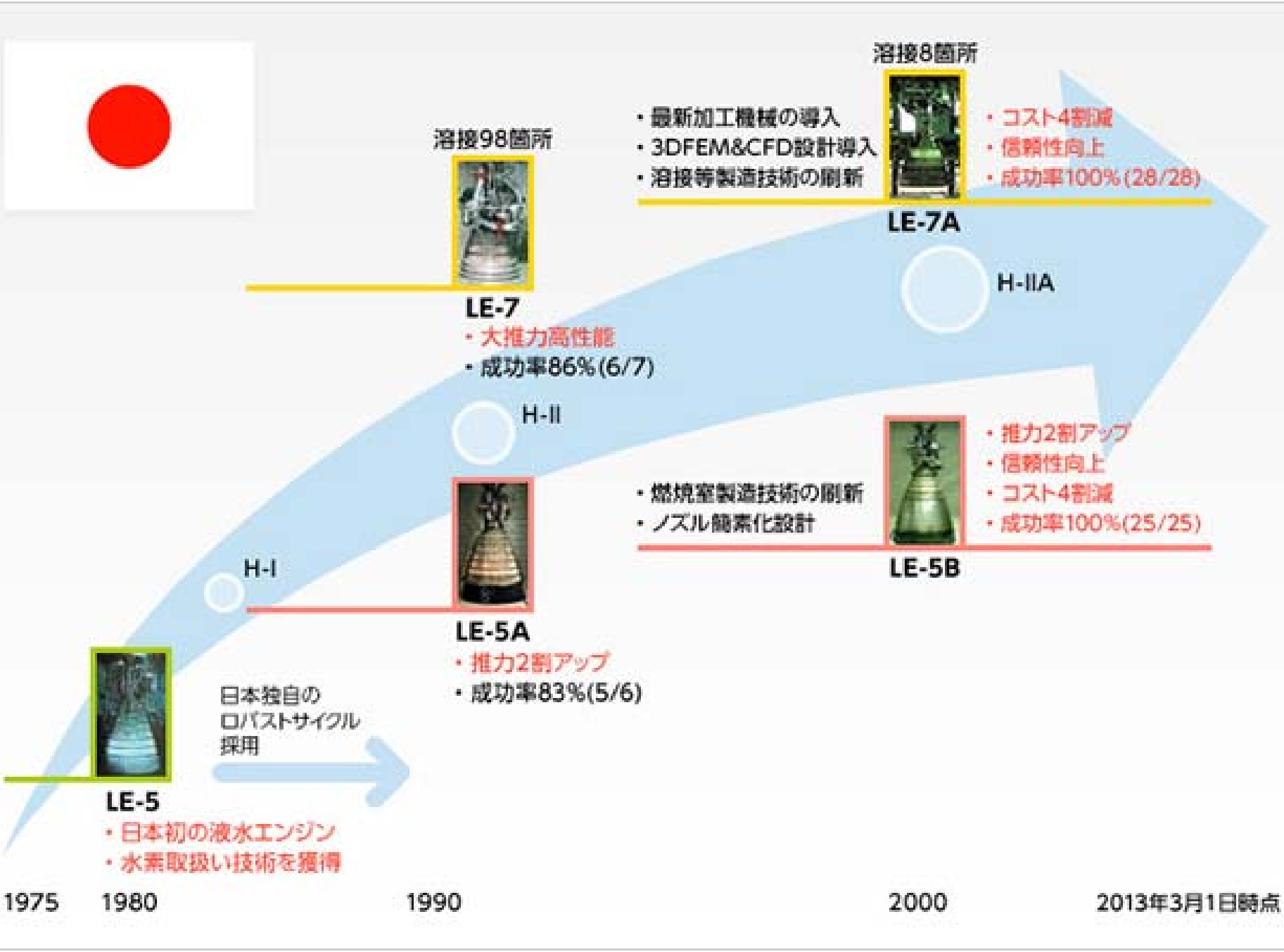
- LE-9の多目的設計探査
  MODE of LE-9

- ロケット射点の空力音響多目的設計探査
  MODE of rocket launch site

LE-9の多目的設計最適化

MODE of LE-9 engine




溶接8箇所
・最新加工機械の導入
・3DFEM&CFD設計導入
・溶接等製造技術の刷新
・コスト4割減
・信頼性向上
・成功率100%(28/28)
LE-7A
溶接98箇所
LE-7
・大推力高性能
・成功率86%(6/7)
H-IIA
H-II
・燃焼室製造技術の刷新
・ノズル簡素化設計
・推力2割アップ
・信頼性向上
・コスト4割減
・成功率100%(25/25)
LE-5B
H-I
LE-5A
・推力2割アップ
・成功率83%(5/6)
日本独自の
ロバストサイクル
採用
LE-5
・日本初の液水エンジン
・水素取扱い技術を獲得
1975
1980
1990
2000
2013年3月1日時点

H3ロケット H3 rocket

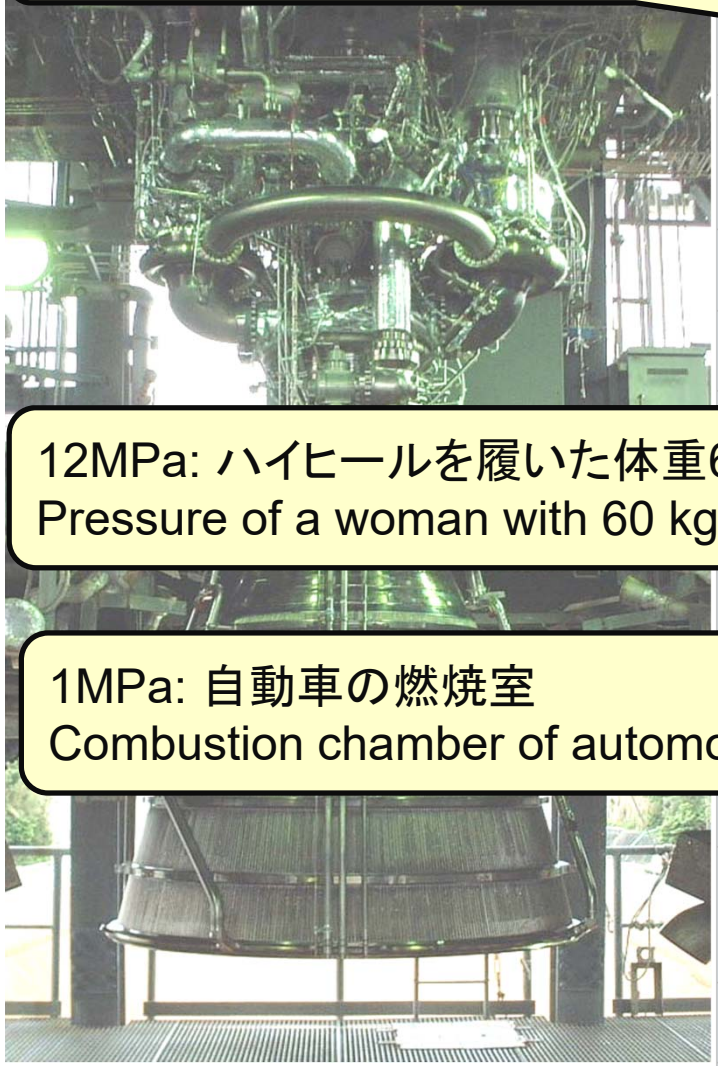
- 日本で開発中の次期基幹ロケット
  H3 rocket is an expendable launch system under development in Japan
- 固体ロケットブースターが就いた液体燃料ロケット
  It is a liquid-propellant rocket with solid rocket boosters
- MHIとJAXAが設計，製造，運用を行う
  MHI and JAXA are responsible for the design, manufacture, and operation.
- 最小構成で4トンのペイロードを太陽同期軌道に50億円で，完全構成で6.5トンのペイロードを静止軌道に運ぶことが出来る
  Minimum configuration carries a payload of up to 4 tons into sun-synchronous orbit for about 5 billion yen, and its maximum configuration carries more than 6.5 tons into geostationary transfer orbit
- 最初のH3ロケットは2020年に打ち上げ予定
  The first H3 rocket is to be launched in 2020.

Current engine LE-7A for H-II A/B rockets

GE90（Boeing777のエンジン）の約2倍
Double of GE90 (engine of Boeing 777)

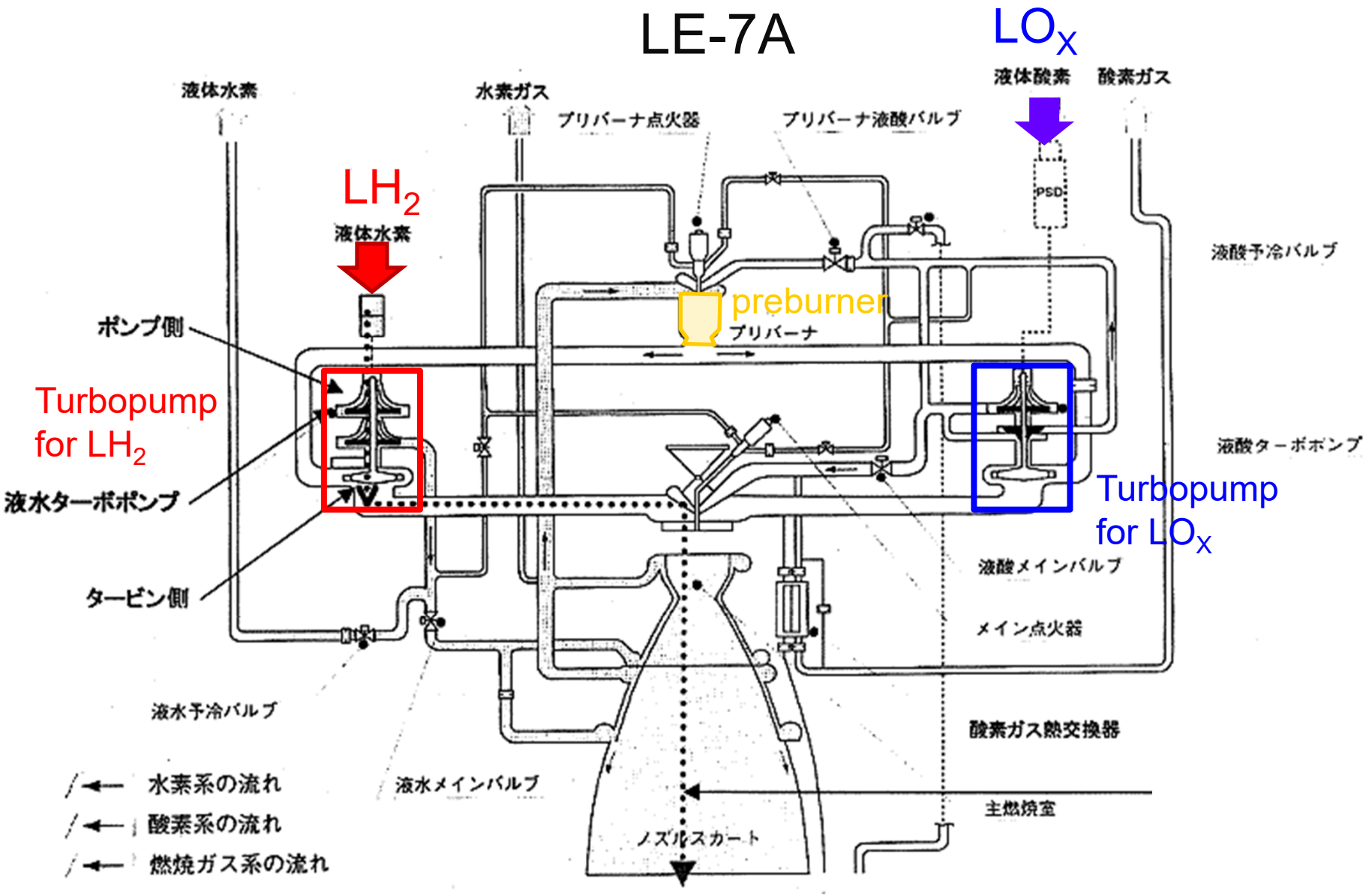
12MPa: ハイヒールを履いた体重60kgの女性による圧力に相当
Pressure of a woman with 60 kg weight wearing high heels

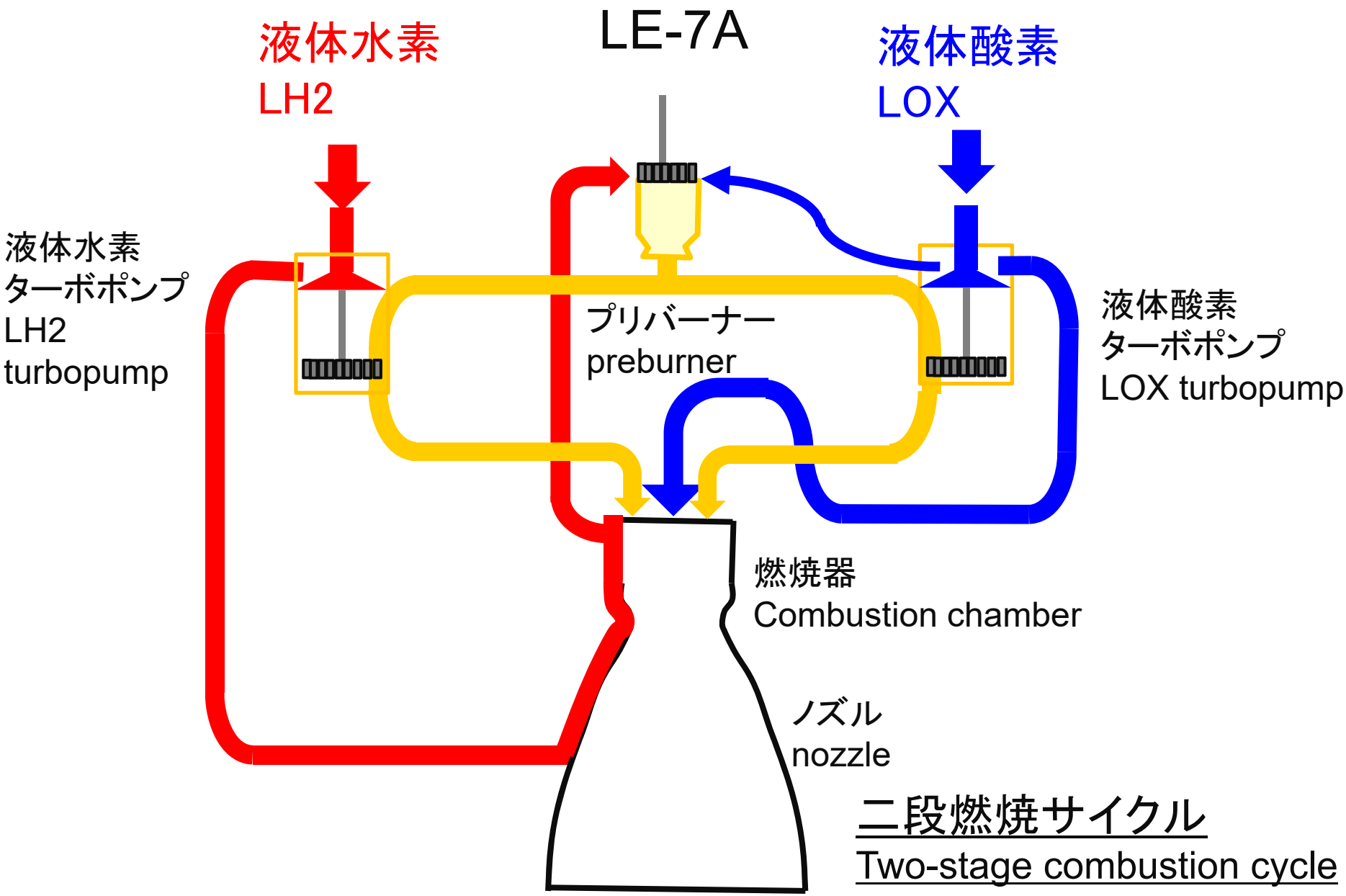
1MPa: 自動車の燃焼室
Combustion chamber of automobile



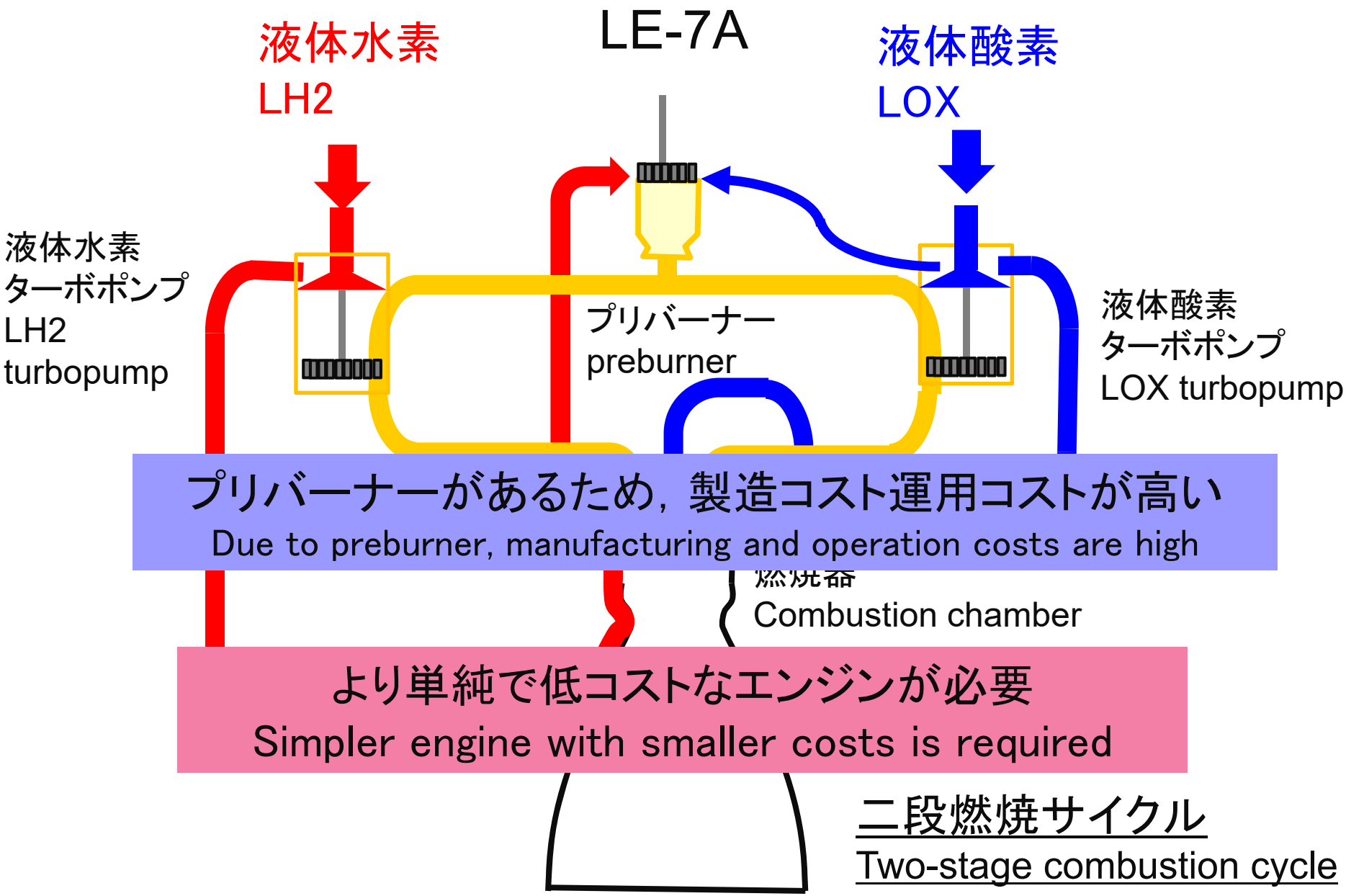
	LE-7	LE-7A	
		Short nozzle	Long nozzle
適用ロケット rocket	H-II	H-IIA	H-IIA/B
真空中推力 Thrust in vacuum	1,079kN (110ton)	1,074kN (109.5ton)	1,098kN (112ton)
混合比 Mixing ratio	6.0	5.9	5.9
Specific impulse	446s	429s	440s
	二段燃焼サイクル		
	LH2/LOX		
Expansion ratio	51.6	約38.7	約46.7
		12.3MPa	
bopump Rotation speed	42,200 回転/min	700回転/秒	700回転/秒
LOX turbopump Rotation speed	18,100 回転/min	18,300 回転/min	18,300 回転/min
全長 length	3.2m	3.4m	3.7m
質量 mass	約1.7ton	約1.7ton	約1.8ton

LE-7A


液体水素
水素ガス
プリバーナ点火器
プリバーナ液酸バルブ
液体酸素
酸素ガス
PSD
液酸予冷バルブ
液体水素
preburner
プリバーナ
ポンプ側
Turbopump for LH2
液水ターボポンプ
タービン側
液酸ターボポンプ
Turbopump for LOX
液酸メインバルブ
メイン点火器
液水予冷バルブ
酸素ガス熱交換器
液水メインバルブ
主燃焼室
ノズルスカート
水素系の流れ
酸素系の流れ
燃焼ガス系の流れ



2段燃焼サイクルの欠点は？ Drawback of TSCC?



プリバーナーがあるため，製造コスト運用コストが高い

Due to preburner, manufacturing and operation costs are high

より単純で低コストなエンジンが必要

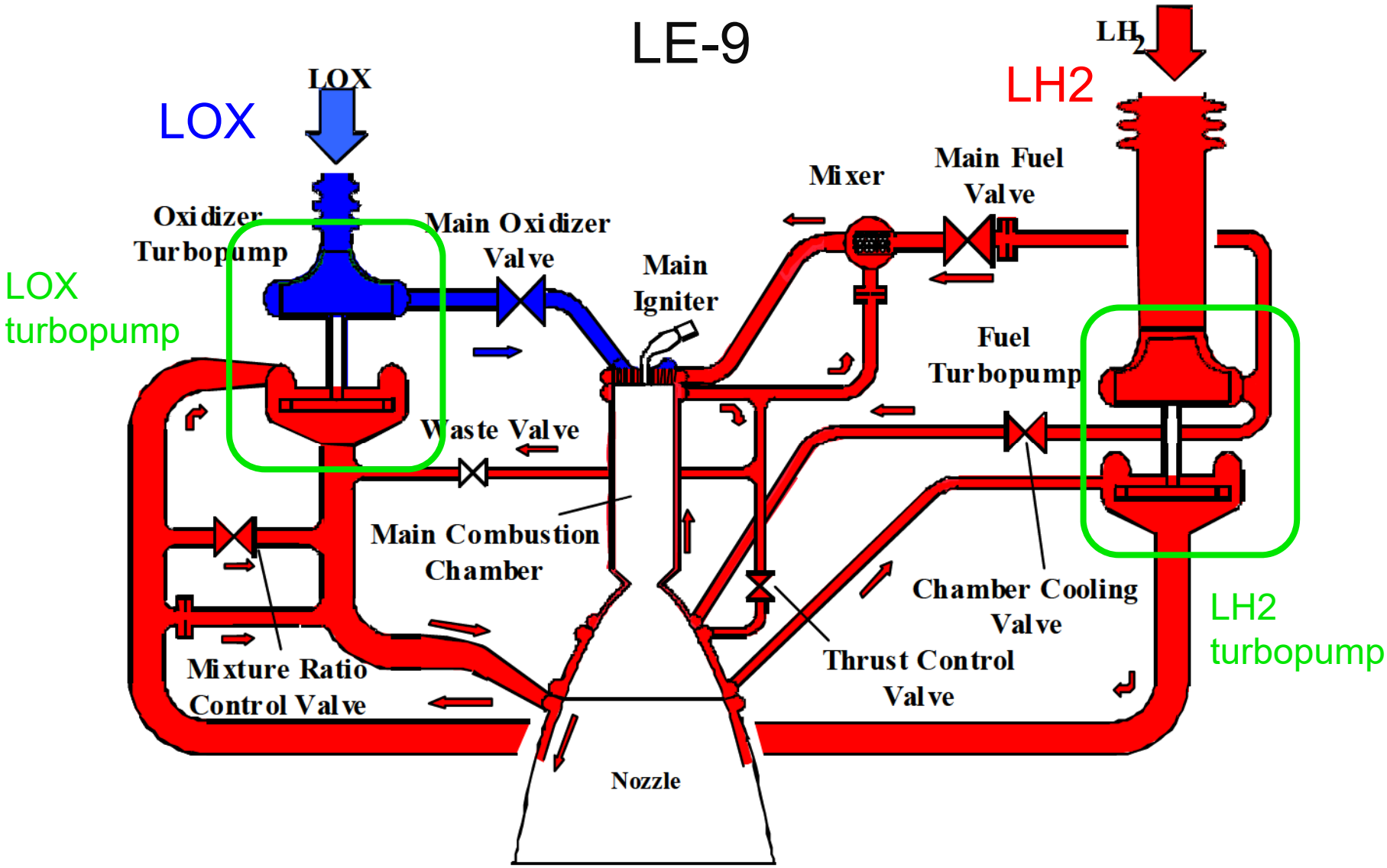
Simpler engine with smaller costs is required

二段燃焼サイクル

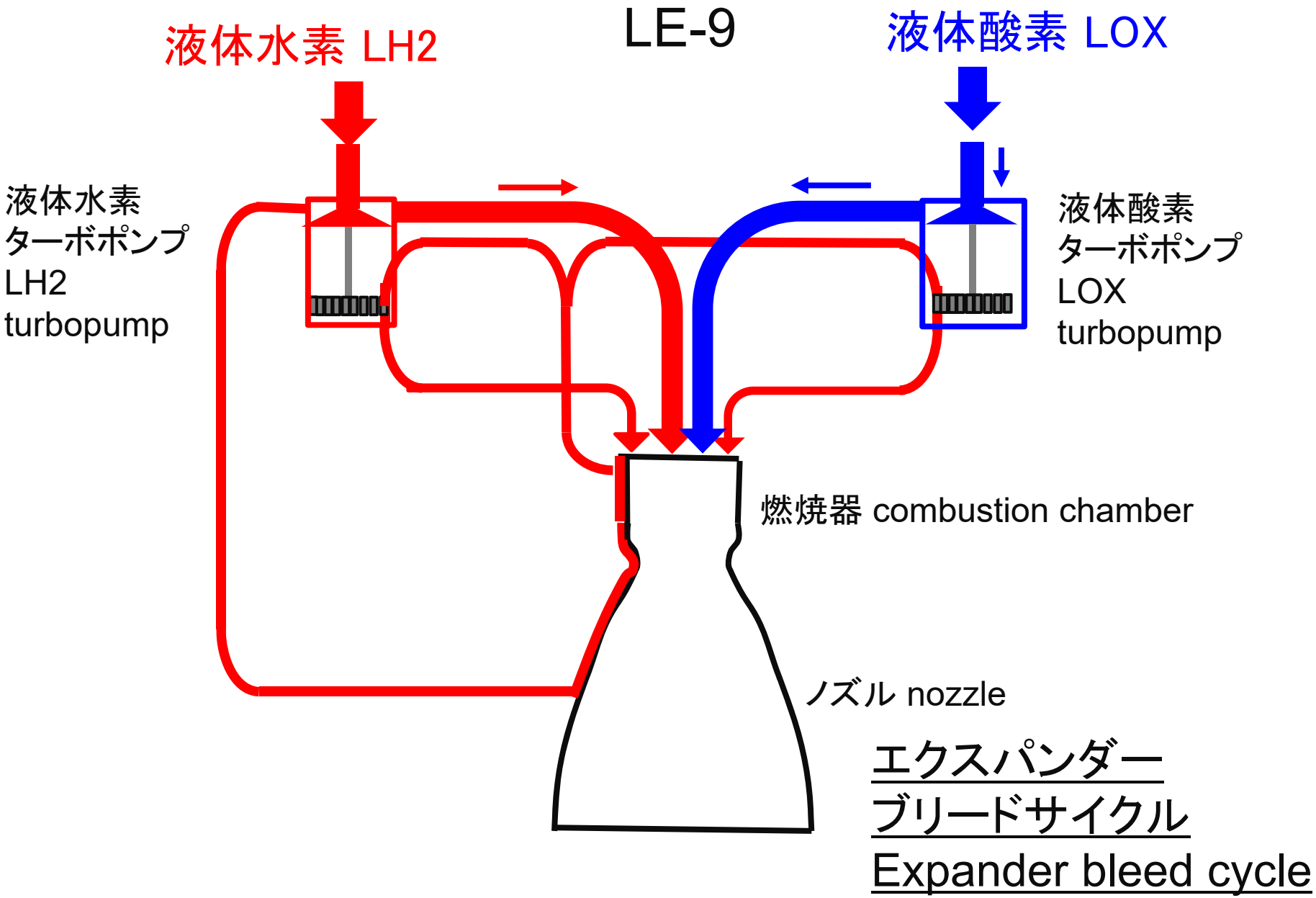
Two-stage combustion cycle

2段燃焼サイクルの欠点は？ Drawback of TSCC?

LE-9


LOX
LOX
LH2
LH2
Oxidizer Turbopump
Main Oxidizer Valve
Mixer
Main Fuel Valve
Main Igniter
LOX turbopump
Fuel Turbopump
Waste Valve
Main Combustion Chamber
Chamber Cooling Valve
LH2 turbopump
Mixture Ratio Control Valve
Thrust Control Valve
Nozzle

LE-9



エクスパンダー
ブリードサイクル
Expander bleed cycle

なぜエクスパンダーブリードサイクル？　Why EBC?

Expander-bleed cycle

- 燃焼前の液体燃料を高温の燃焼室やノズルの周囲へ巡らして冷却すると共に熱交換で気化した燃料でターボポンプを駆動する

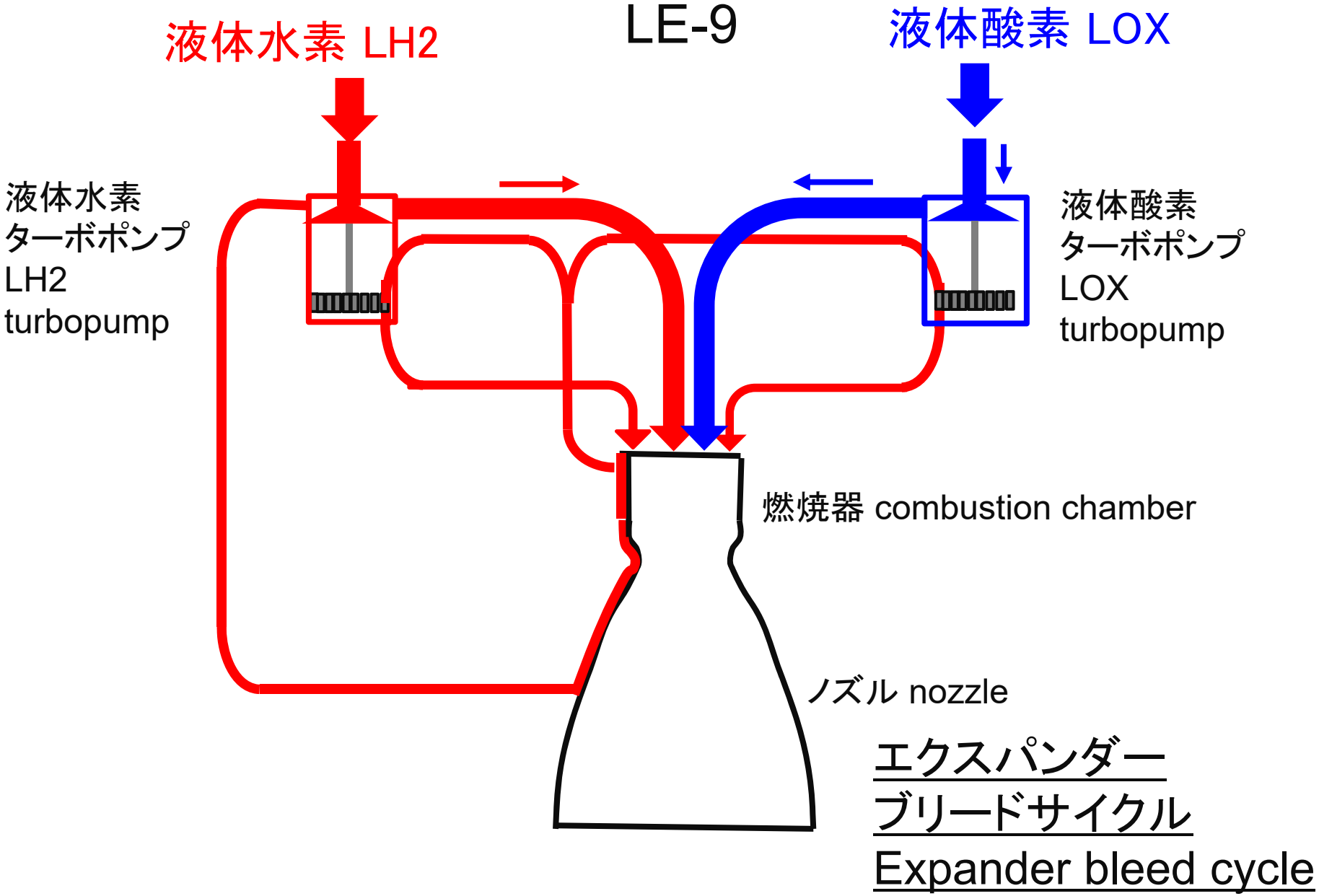
  Cool down combustion chamber and nozzle with the liquid fuel and the vaporized fuel drive the turbo pump



- プリバーナーがないため，製造コスト運用コストが安価

  Low manufacturing and operation costs because there is no preburner

LE-9



エクスパンダー
ブリードサイクル
Expander bleed cycle

2段燃焼サイクルに比べて技術的に難しい点は？ difficulty?

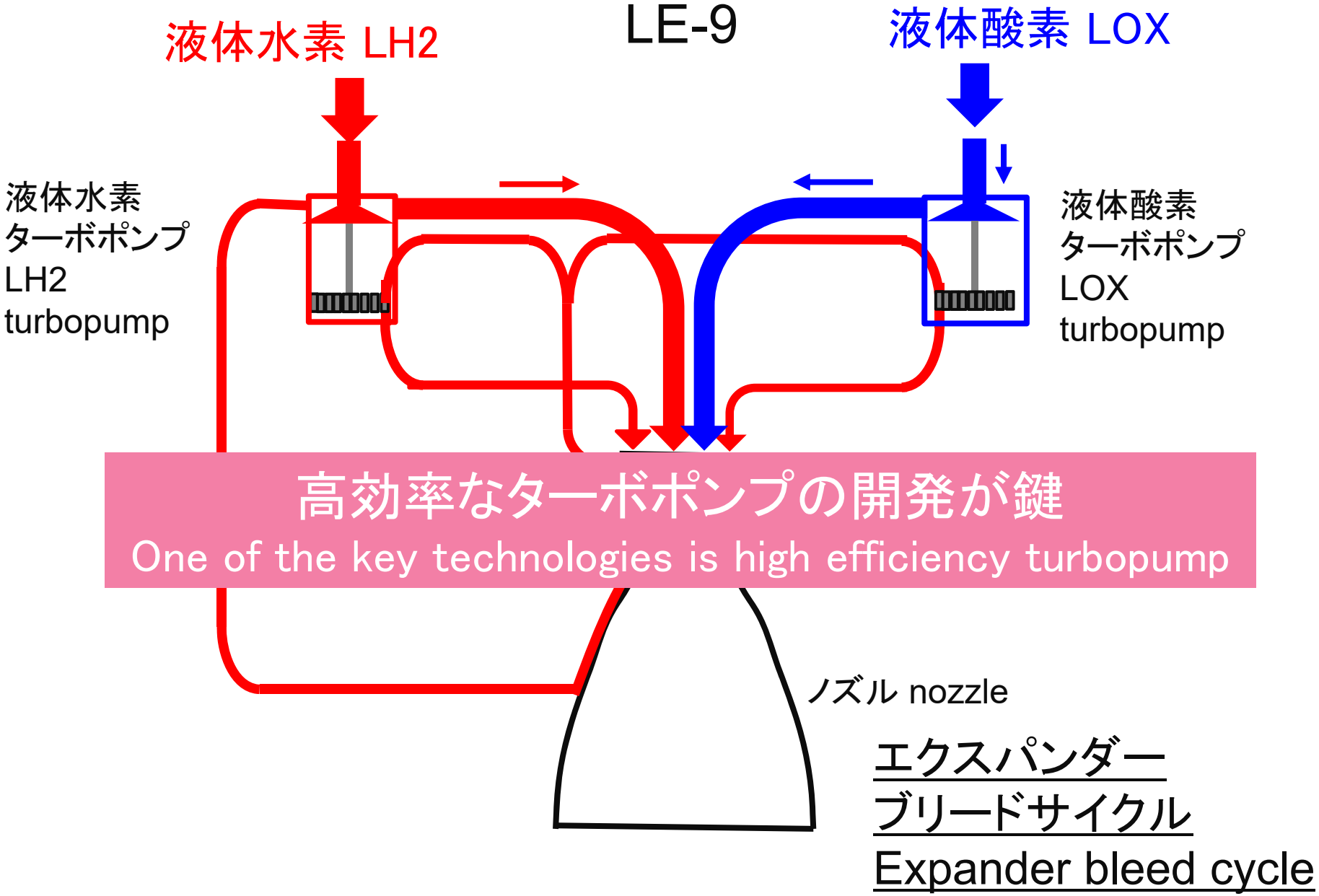
LE-9

	LE-9	LE-7A(参考)
燃焼サイクル	エキスパンダブリードサイクル	二段燃焼サイクル
真空中推力 tf	148	112
混合比	5.9	5.9
真空中比推力 s	430	440
燃焼圧力 MPa	12	12
LH_2ターボポンプ回転数 min^{-1}	40,000	41,900
LOXターボポンプ回転数 min^{-1}	16,500	18,300

ポンプに要求される性能はLE-7Aとほぼおなじ
Required performance is same as LE-7A

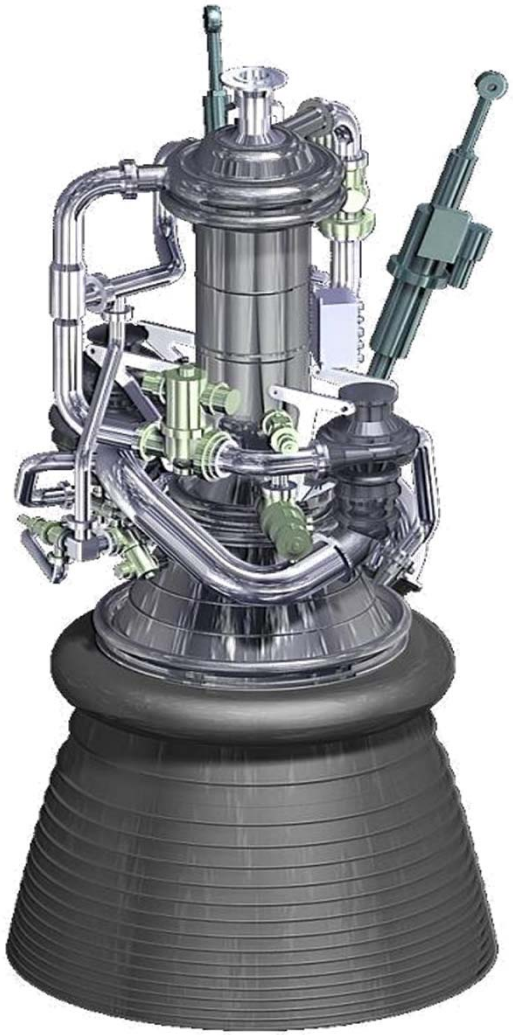
しかしながら，エネルギー源は燃焼室やノズルでの熱交換のみ
However, energy source is heat exchange only in combustion chamber and nozzle

LE-9



2段燃焼サイクルに比べて技術的に難しい点は？ difficulty?

LE-9 engine



燃料(液体水素)流量：毎秒約200リットル
Fuel (liquid hydrogen) flow rate:
200 liters per second

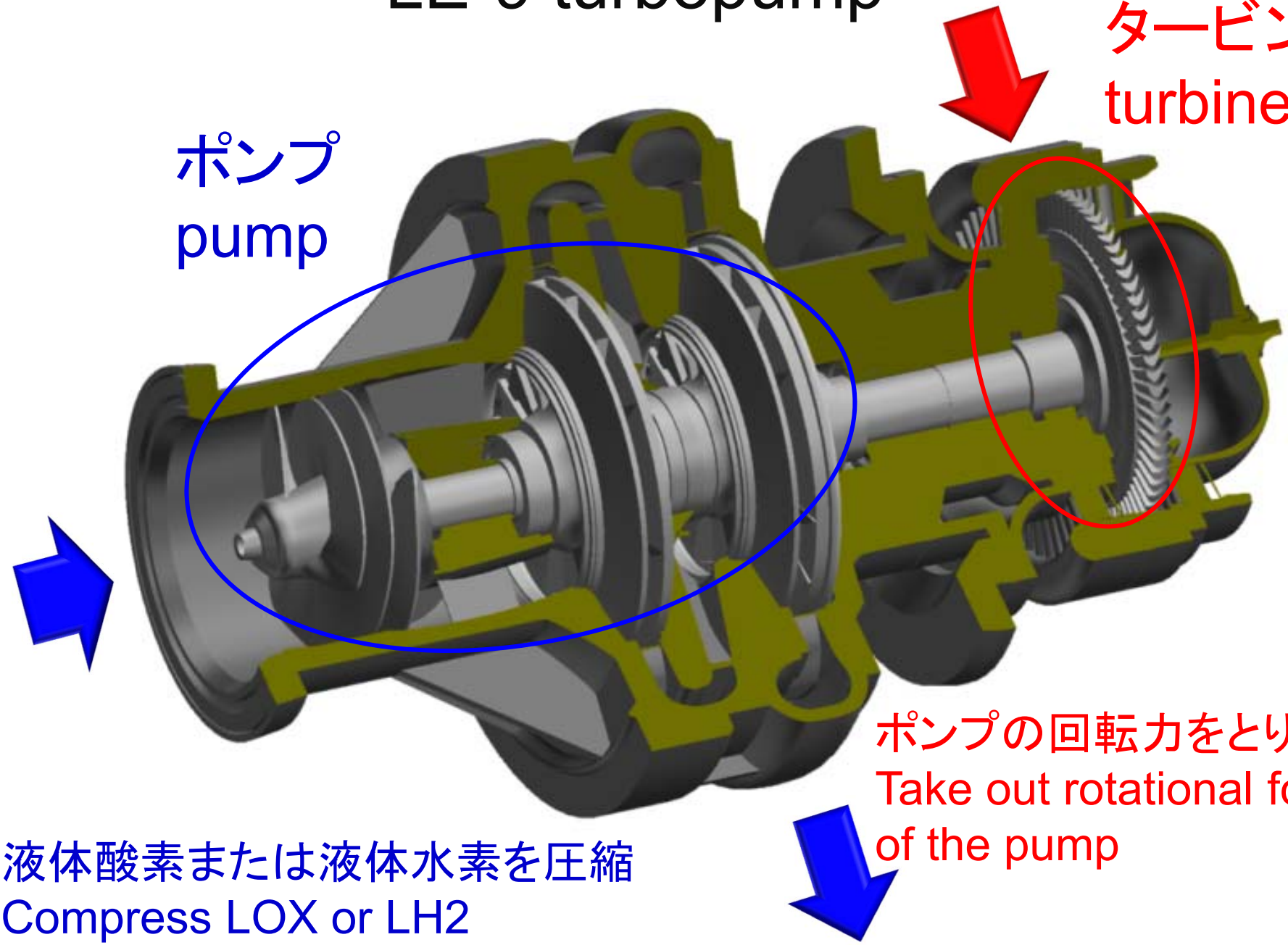
酸化剤(液体酸素)流量：毎秒約700リットル
Oxidant (Liquid oxygen) flow rate:
700 liters per second

燃料・酸化剤を送り込むためのタービンには
燃料のごく一部(10kg/s程度)が流れる
Only a small part of the fuel (about 10 kg / s) flows
in the turbine for feeding fuel and oxidizer

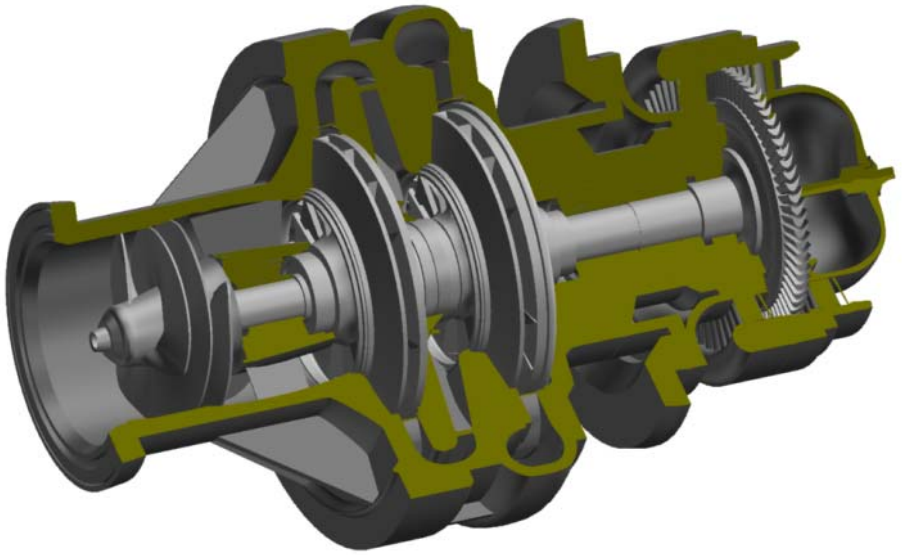
【ミニ知識】LE-9の燃費は？
Fuel consumption rate?

最良の状態でリッター1mm以下
1 mm/litter in the best condition

LE-9 turbopump



LE-9 エンジンのターボポンプ
turbopump for LE-9 engine



燃料側ポンプ pump for fuel

回転数：42,000 rpm (700rps)
必要馬力： 30,000 hp

酸化剤側ポンプ pump for oxidant

回転数：18,000 rpm (300rps)
必要馬力： 6,000 hp

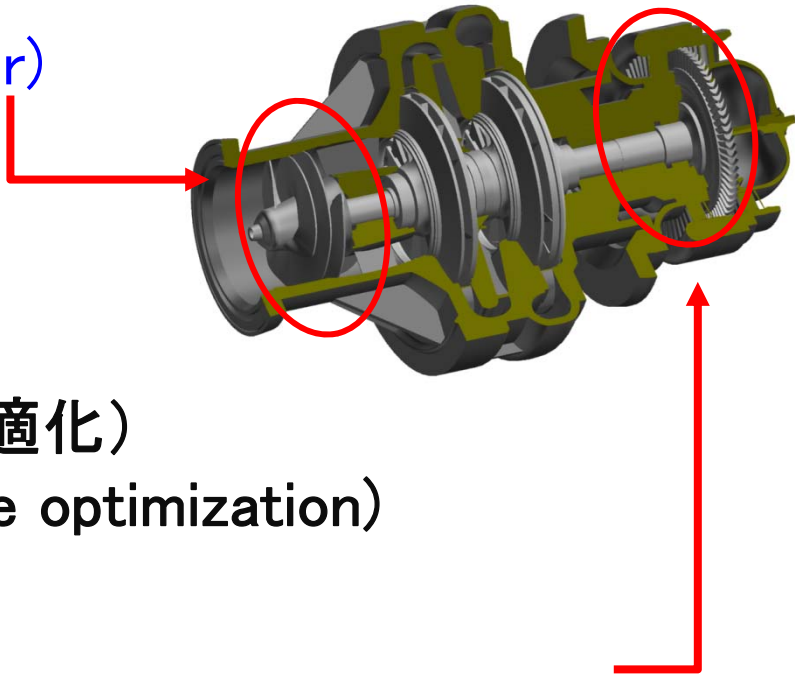
Case1 : ポンプインペラブレード設計（単目的最適化）
pump impeller blade design (single objective optimization)

目的 :必要軸馬力の最小化

objective: minimization of required power)

制約:?

Constraint: ?



Case 2: タービンブレード設計（多目的最適化）
Pump turbine blade design (multiobjective optimization)

目的1:出力軸馬力の最大化

Objective 1: maximization of output power

目的2:?

Objective 2: ?

目的3:?

Objective 3 : ?

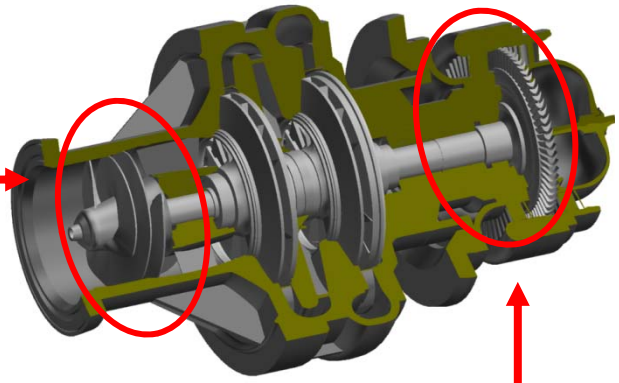
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目的 :必要軸馬力の最小化

objective: minimization of required power)

制約:指定した圧力比になること

Constraint: required head rise is maintained



Case 2: タービンブレード設計（多目的最適化）
Pump turbine blade design (multiobjective optimization)

目的1:出力軸馬力の最大化

Objective 1: maximization of output power

目的2:?

Objective 2: ?

目的3:?

Objective 3 : ?

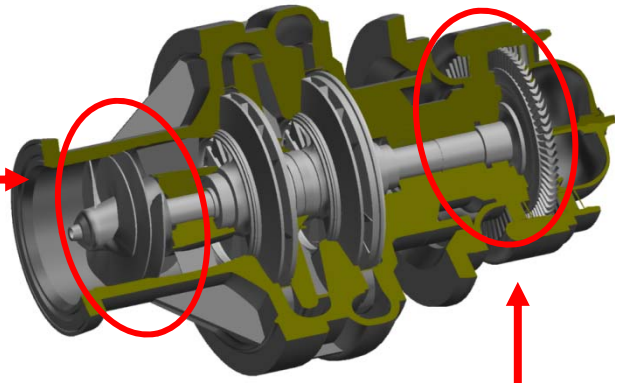
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Case 2: タービンブレード設計（多目的最適化）
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目的1:出力軸馬力の最大化

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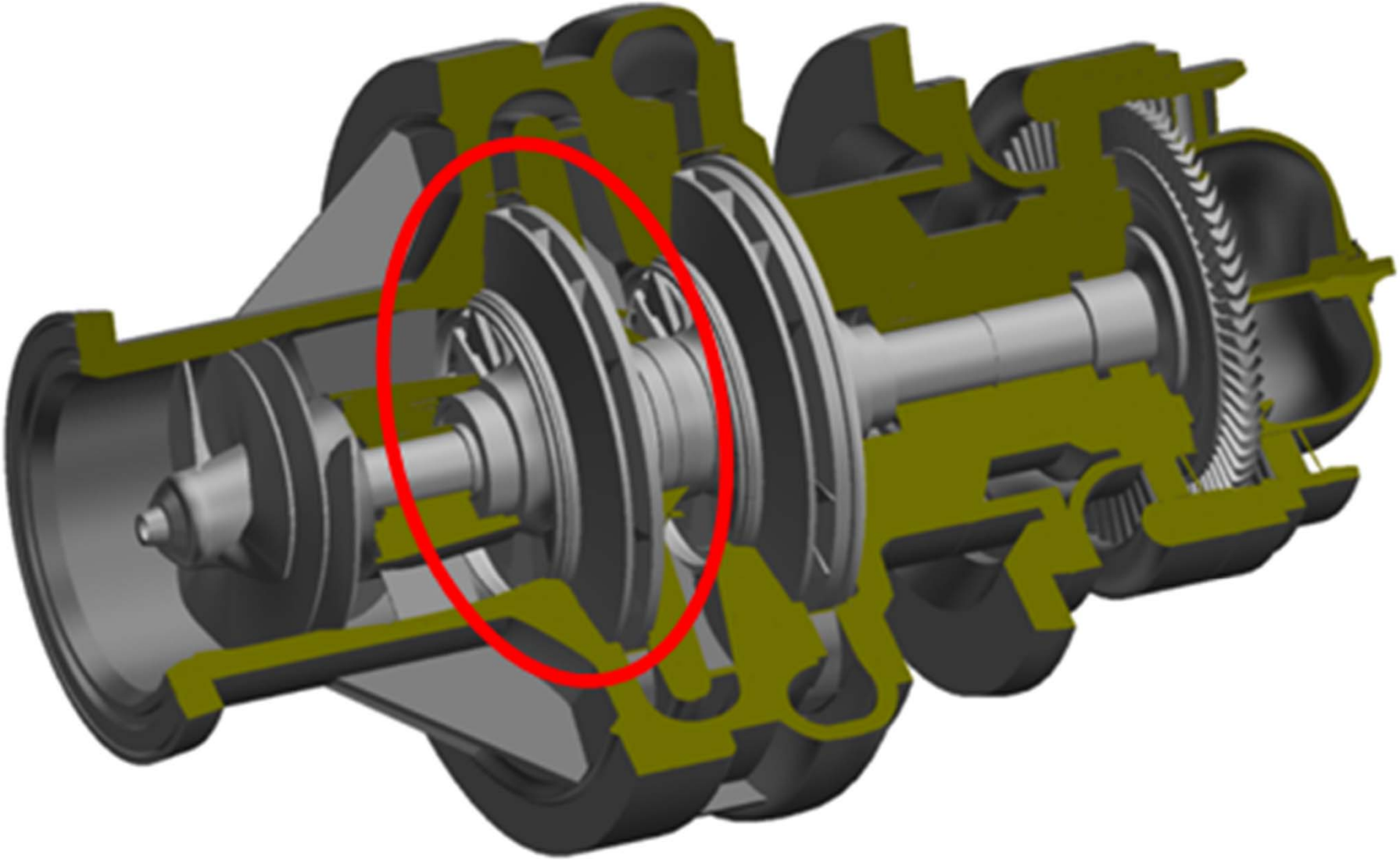
目的2:エントロピー上昇（＝流体エネルギ損失）の最小化

Objective 2: minimization of entropy rise (flow energy loss)

目的3:次段静翼の迎角最小化

Objective 3 : minimization of angle of attack of the next stator

インペラブレードの最適化（単目的最適化）
impeller blade design optimization (single objective)



The Objective Function

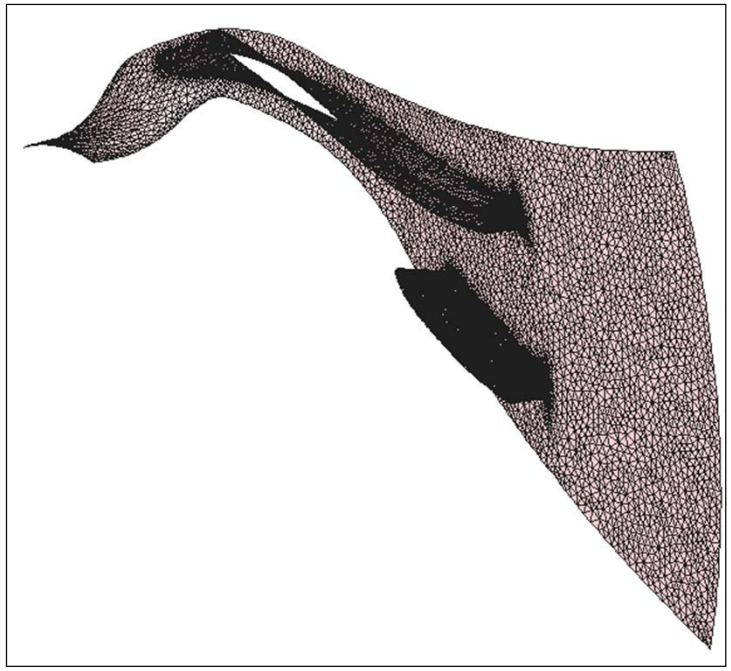
The Objective Function

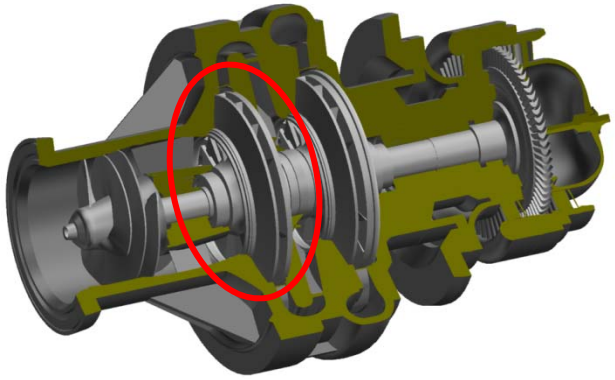
- Shaft horsepower**: Minimize** ↘

The Constraint Function

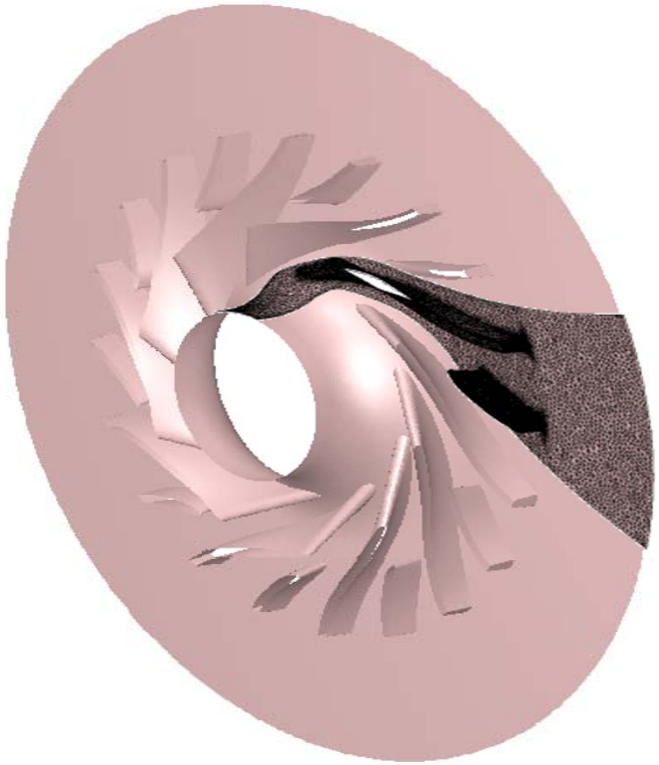
- Impeller Total Head Rise : **Constant**

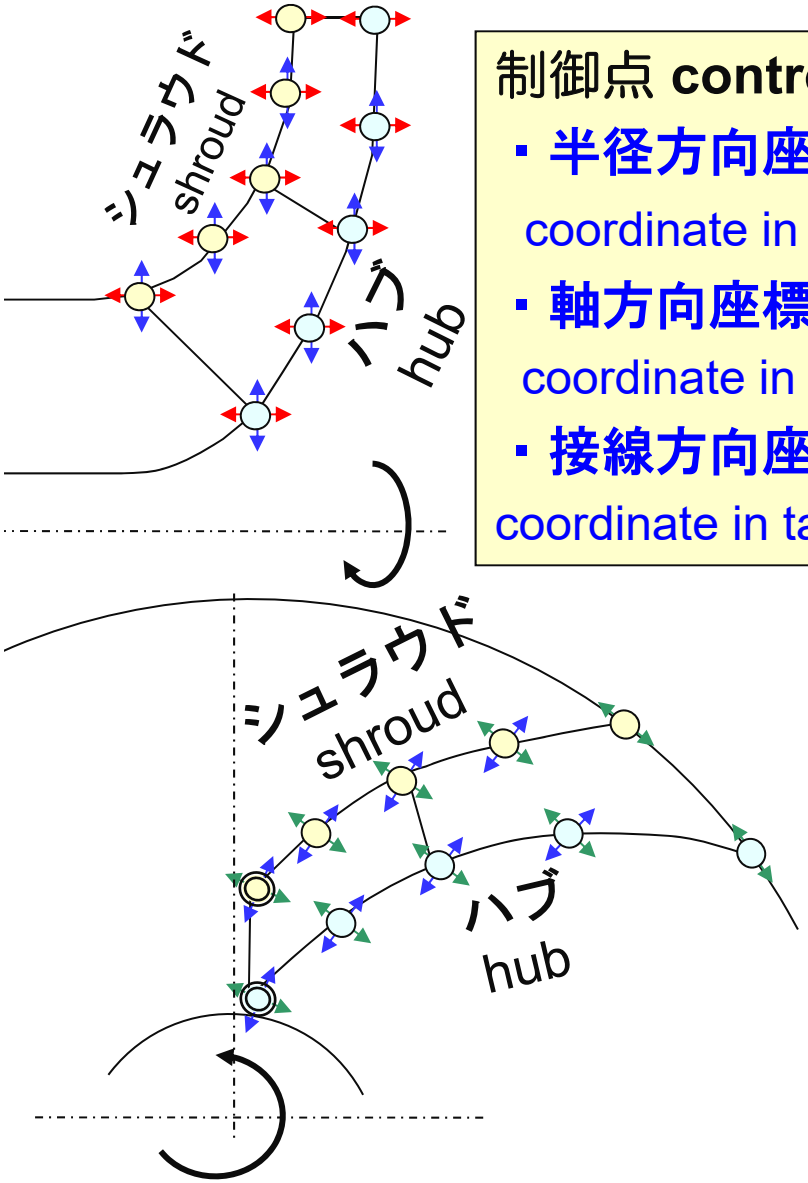






設計変数
design parameters





制御点 **control points**

- **半径方向座標**
  coordinate in radial direction
- **軸方向座標**
  coordinate in axial direction
- **接線方向座標**
  coordinate in tangential direction

合計２８変数
28 design parameters

設計条件 design condition

回転数 (revolution speed): 42,000 rpm
流体 (operating fluid) : 液体水素 (liquid oxygen)

optimization approach

CFD software:

FLUENT 6.3.29

Pressure-Velocity Coupling Method:

Compressible/Incompressible SIMPLE algorithm

Advection Scheme:

Second-order Upwing

Turbulence Model:

Realizable k-e MODEL+Non-equilibrium wall function

Optimization algorithm: single objective evolutionary algorithm

16 Populations x 50 Generations = 800 CFD runs.

Optimization flowchart

CFD grid generation of nominal design

Grid deformation (SCULPTOR)

CFD computation (FLUENT)

Computation of objective function and constraint values

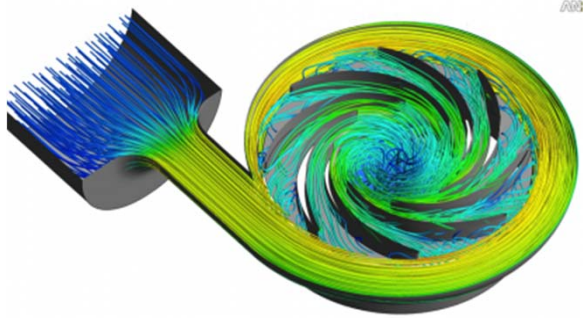
Single objective evolutionary algorithm

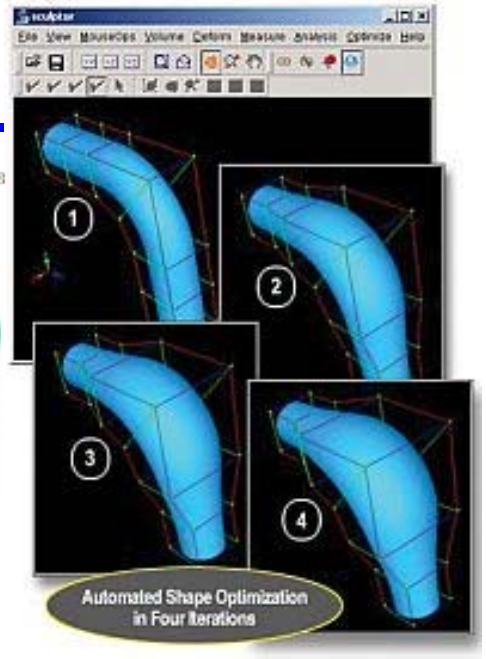
Converged?

NO

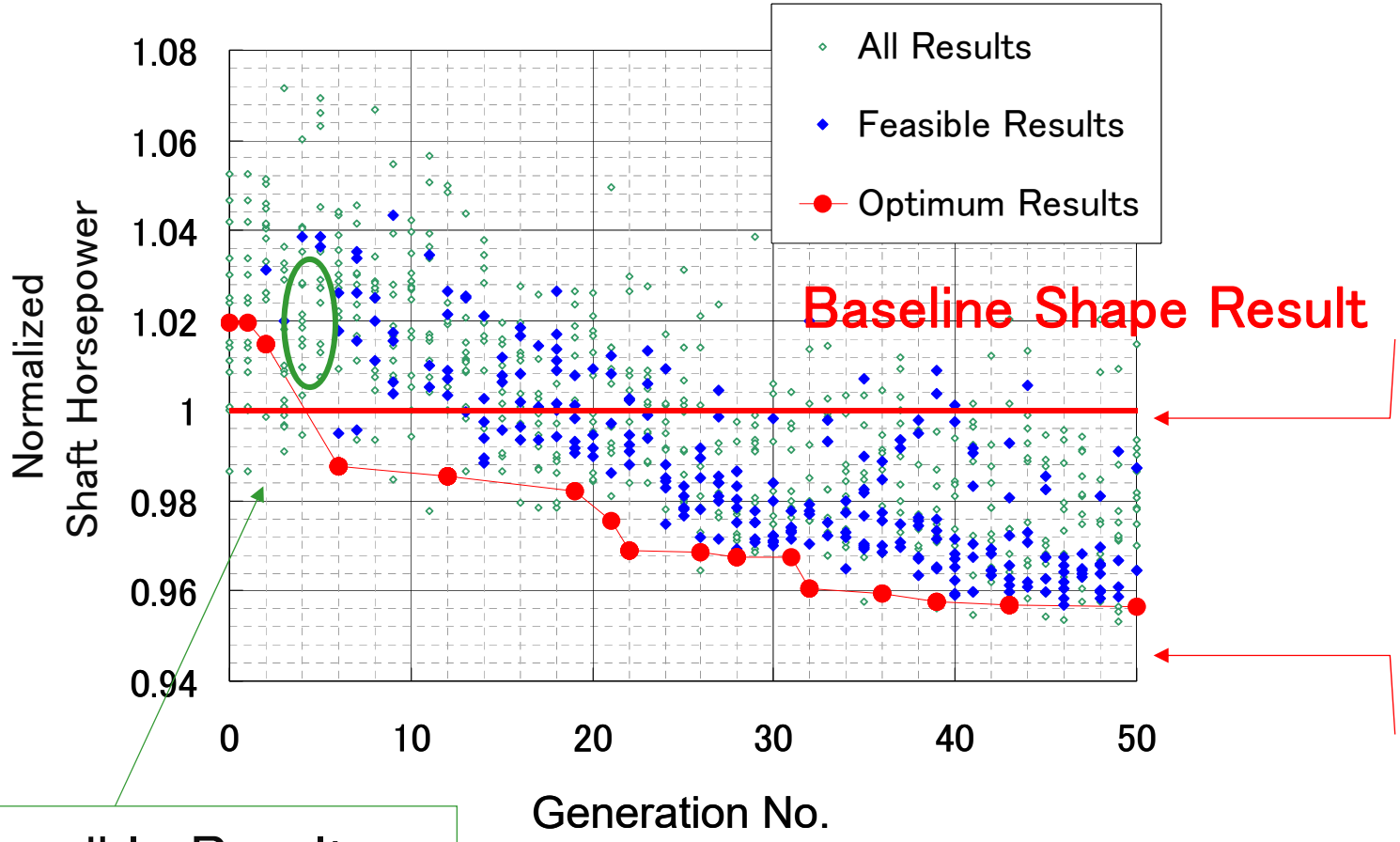
YES

end





Optimization history

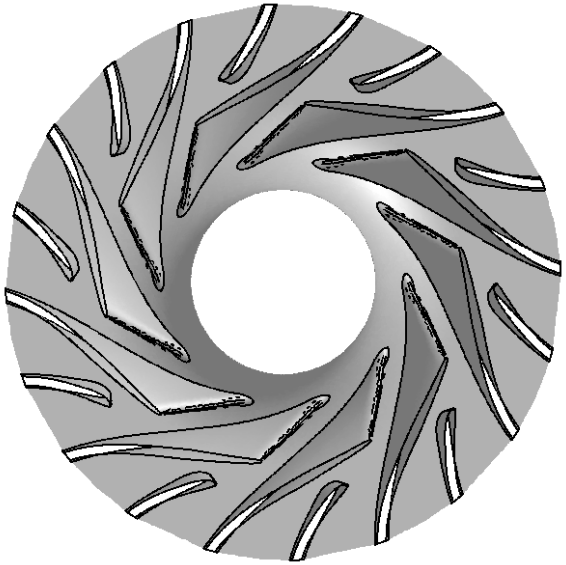


Few Feasible Results

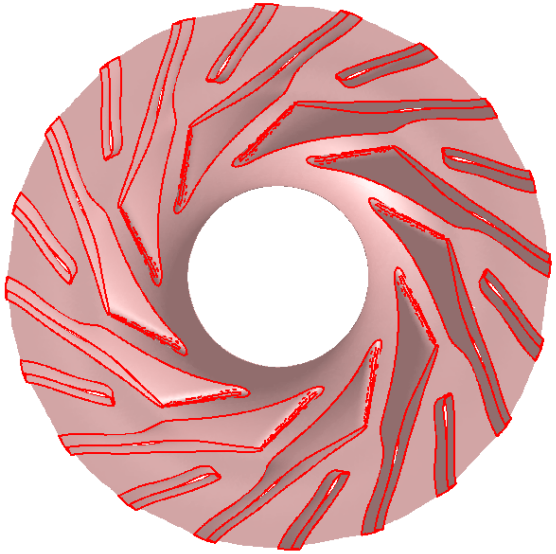
The Final Result
4.4% Lower Shaft Horsepower

The Optimization by Genetic Algorithm is effective!

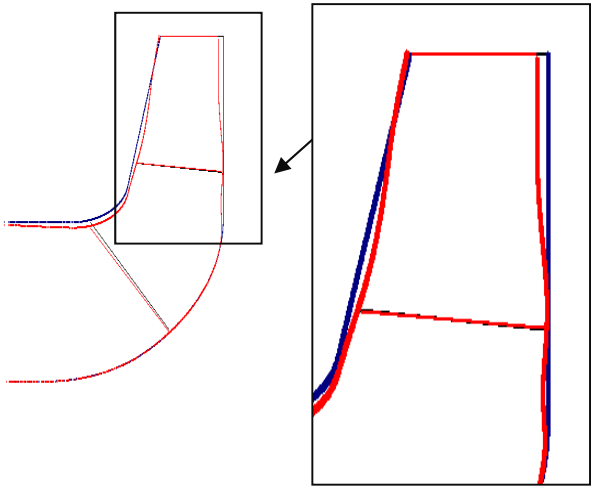
Optimized design

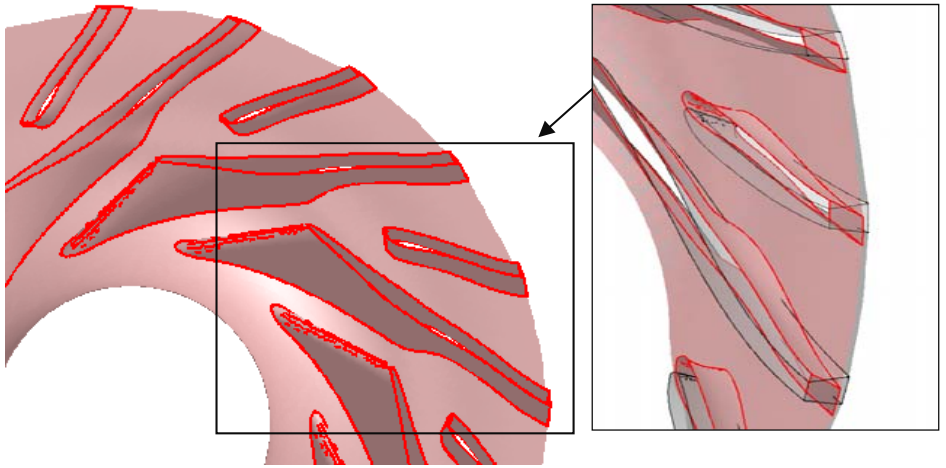


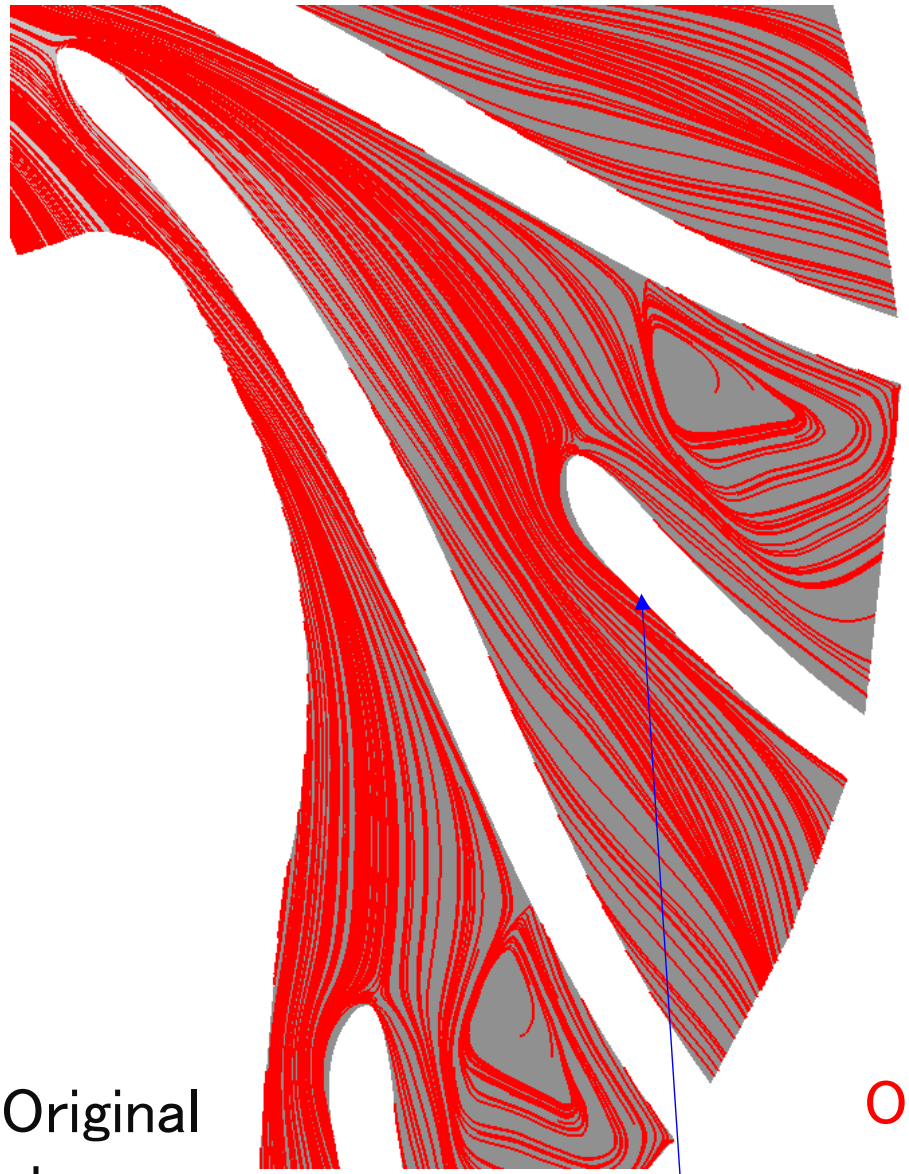
Original shape



Optimized shape



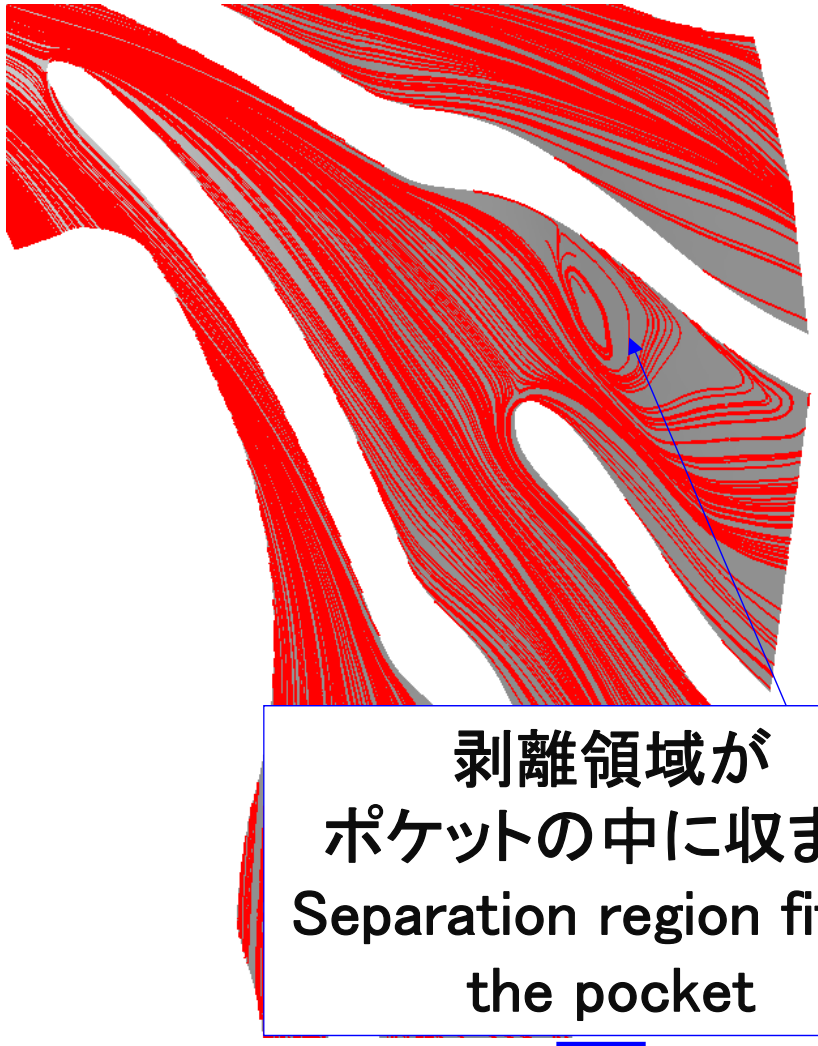




Original shape

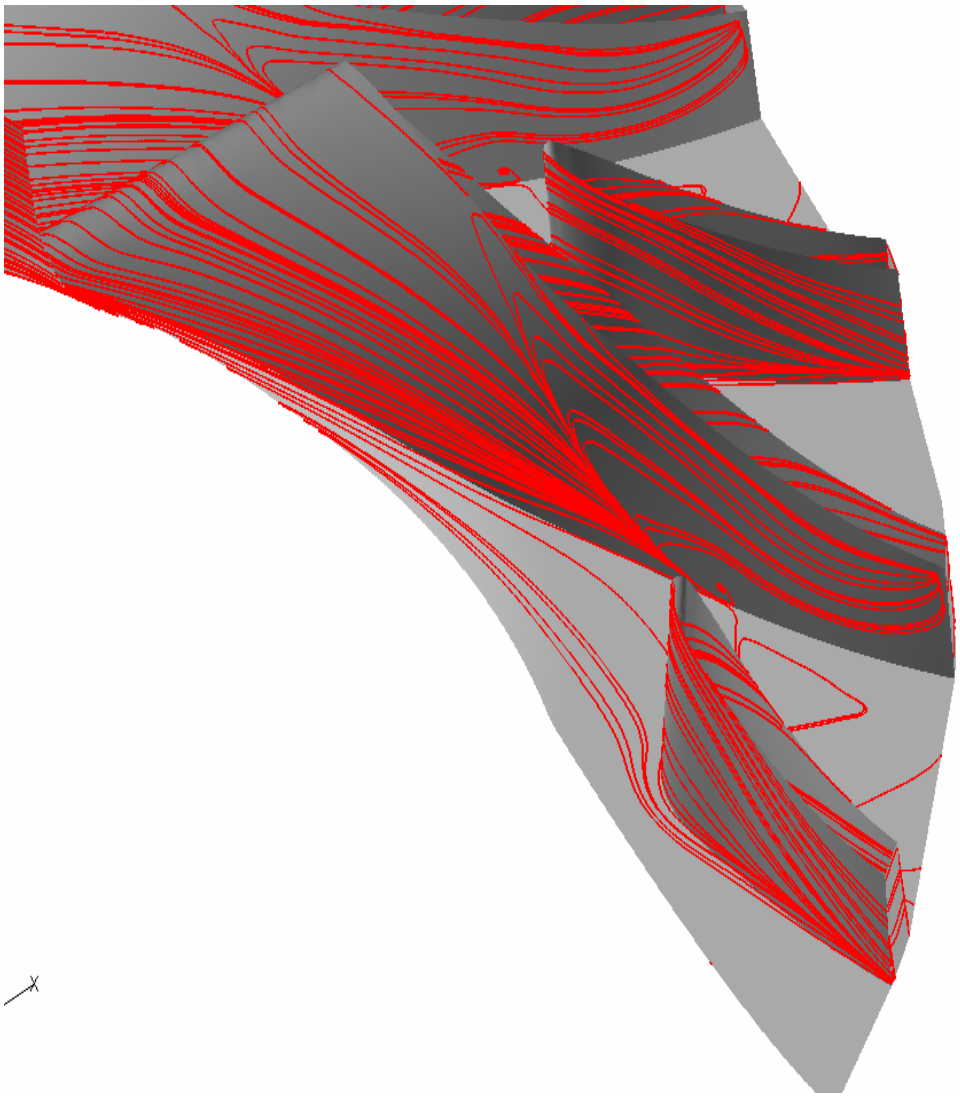
剥離領域が流路をブロック
Separation region block the flow path

Optimized shape

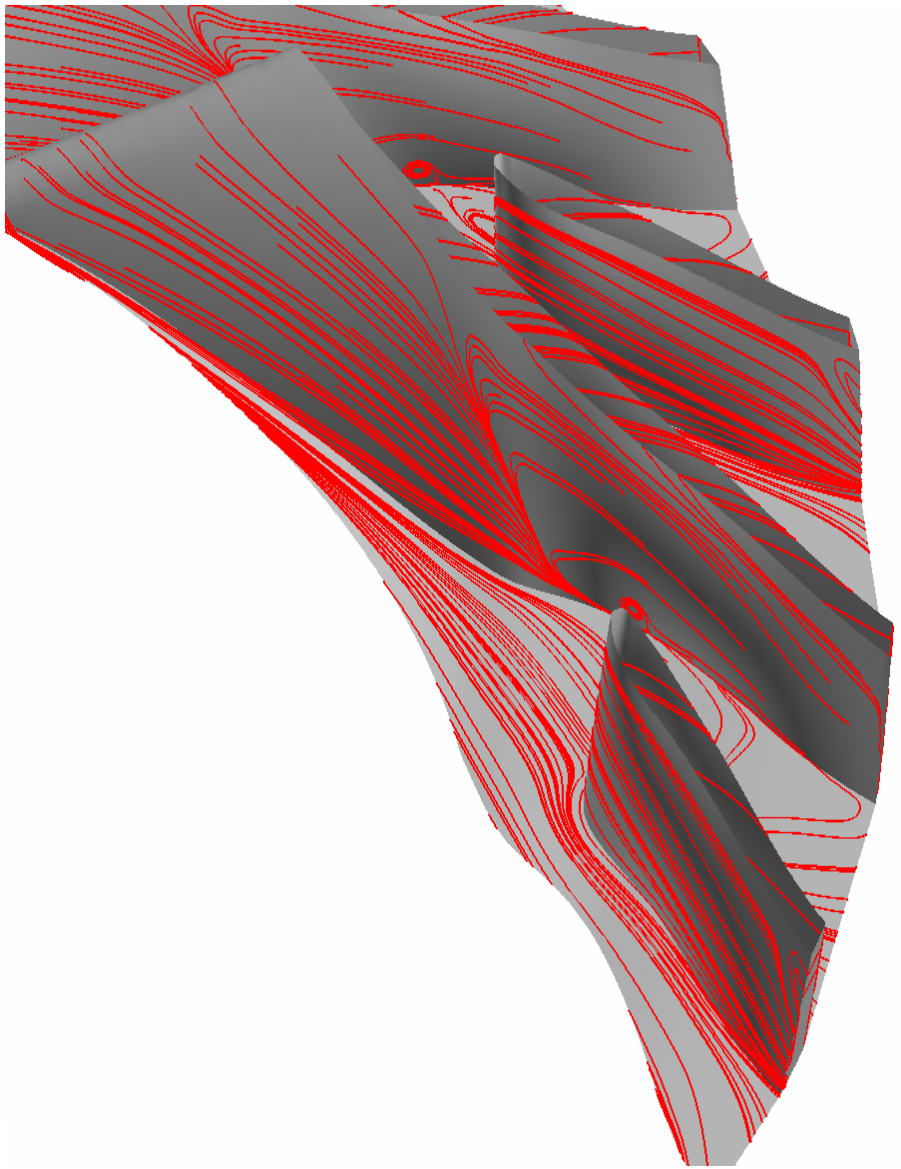


剥離領域が
ポケットの中に収まる
Separation region fits in the pocket

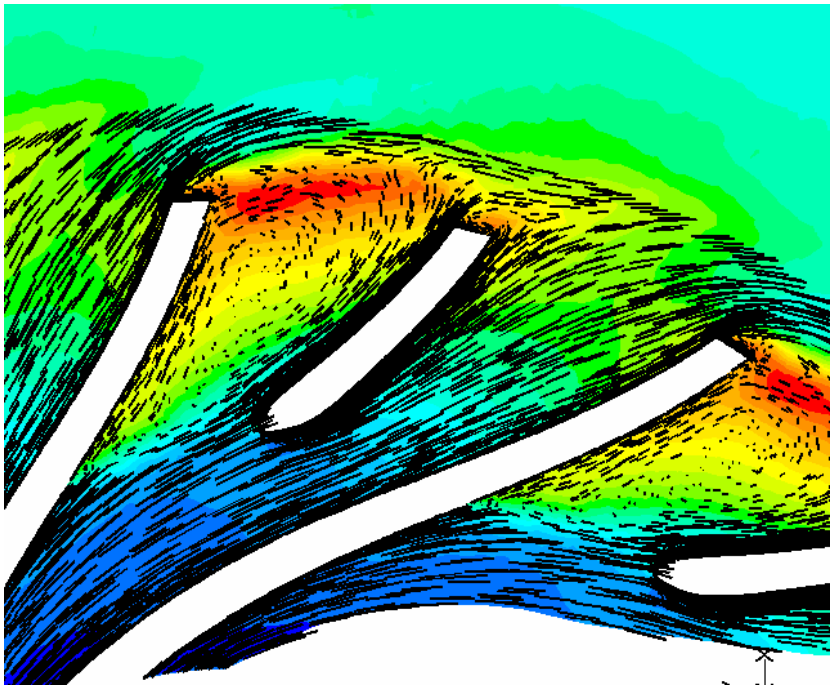
流路が広がる
Wider flow path



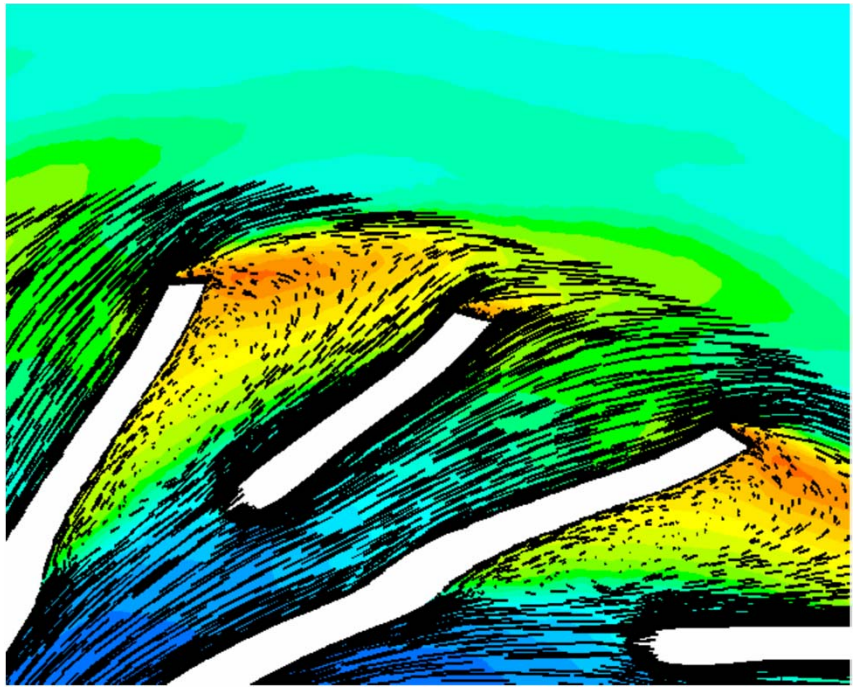
Baseline



Optimum



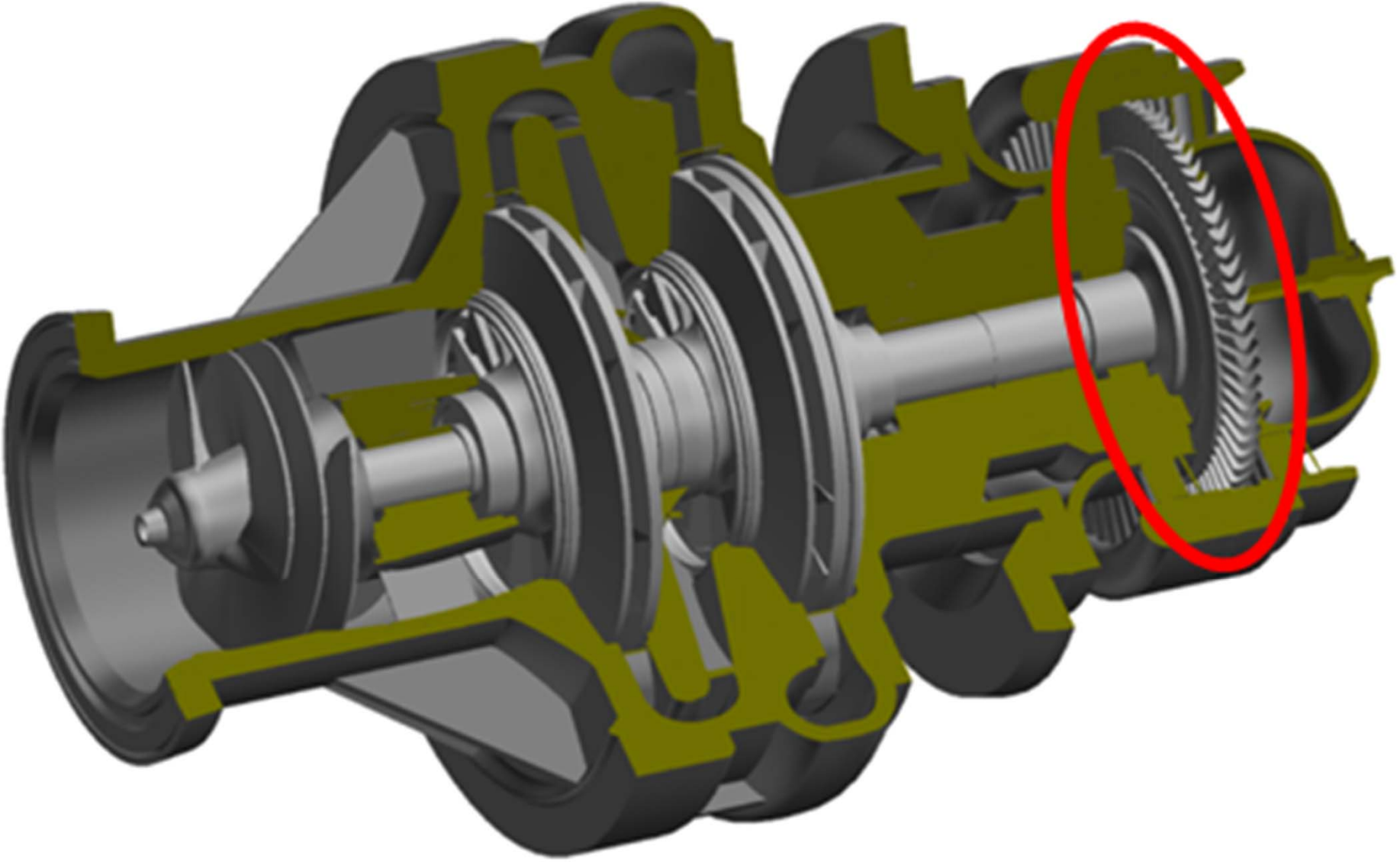
オリジナル形状の動圧分布
Dynamic pressure distribution
of original design

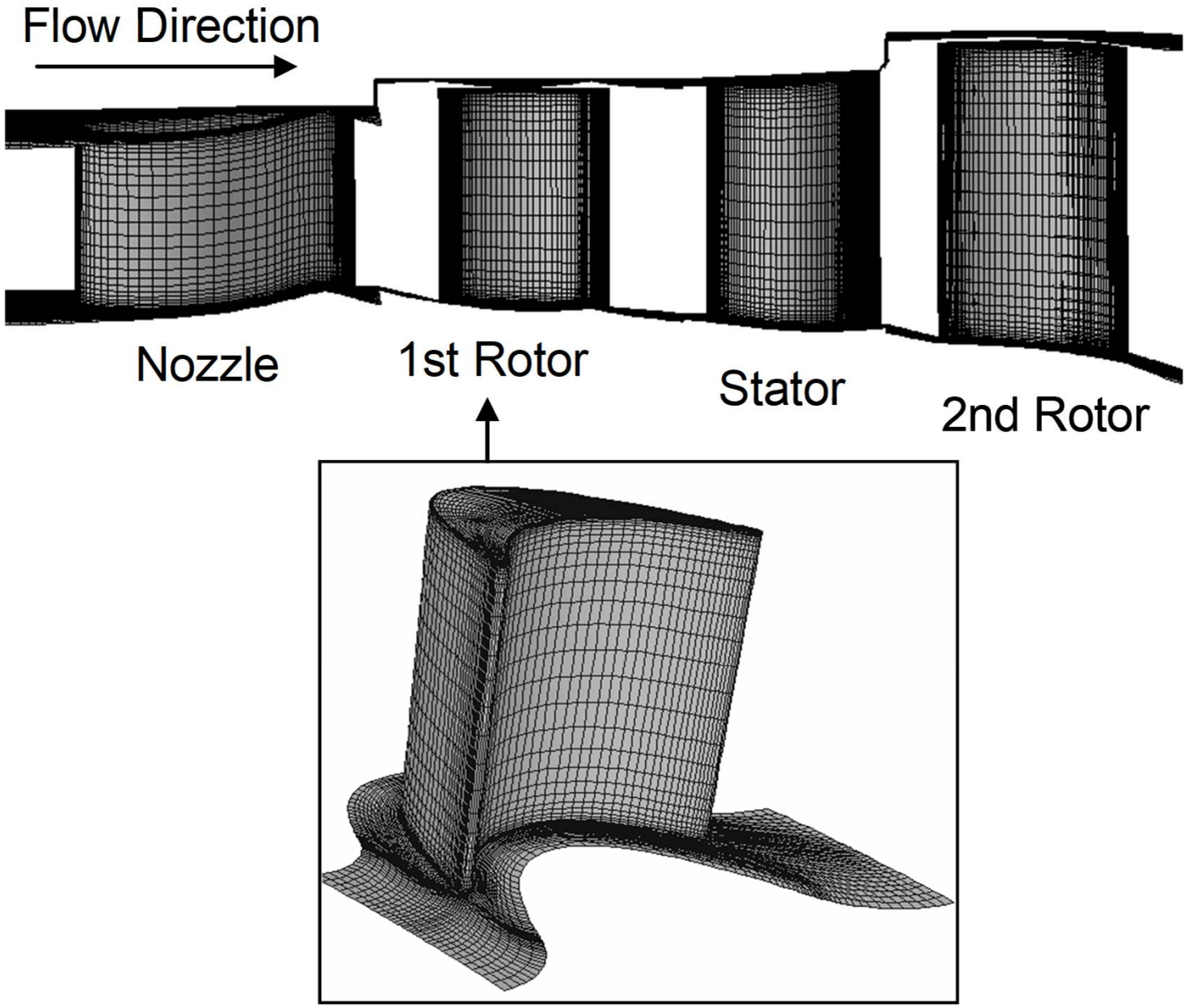


最適形状の動圧分布
Dynamic pressure distribution
of optimized design

流体ロスにつながる動圧が低くなっている
Dynamic pressure leading to fluid loss is lower

タービンブレードの多目的設計最適化
turbine blade multiobjective design optimization





最もエントロピー上昇が大きい1段動翼を最適化の対象とする

First rotor is optimized because entropy rise is largest

ケース2：タービンブレード設計（多目的）
Case 2 : Multiobjective design optimization of turbine blade

目的1：出力軸馬力の最大化

Objective 1: maximization of produced power

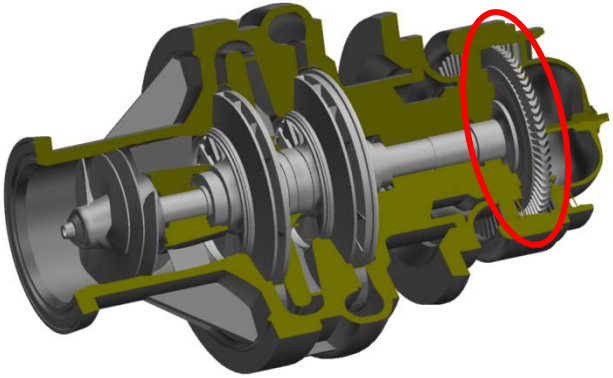
目的2：エントロピー上昇（＝流体エネルギ損失）の最小化

Objective 2 : minimization of entropy rise

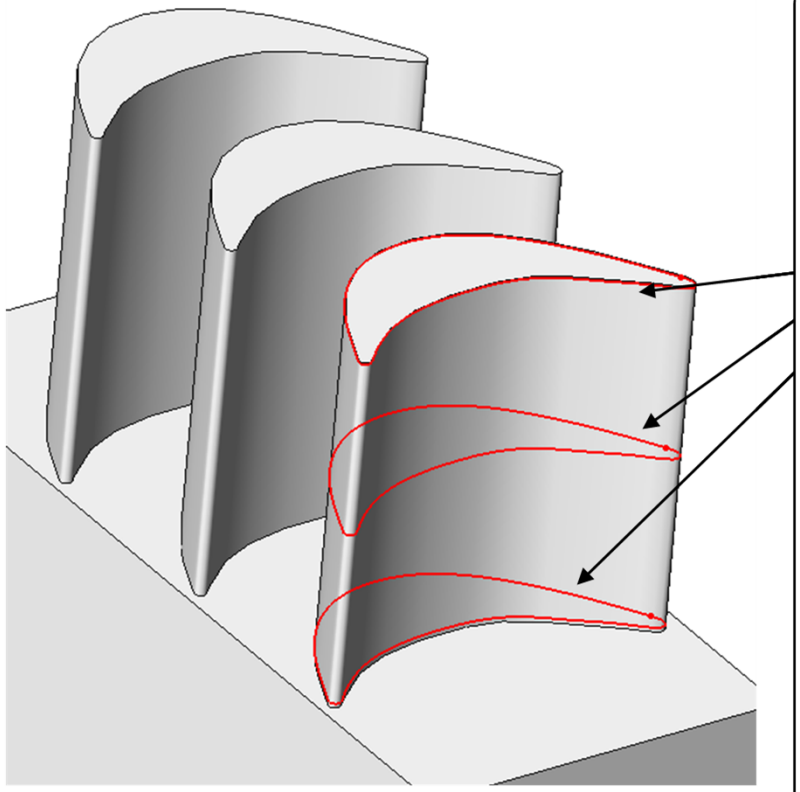
(flow loss minimization)

目的3：次段静翼の迎角最小化

Objective 3: minimization of angle of attack of next stator

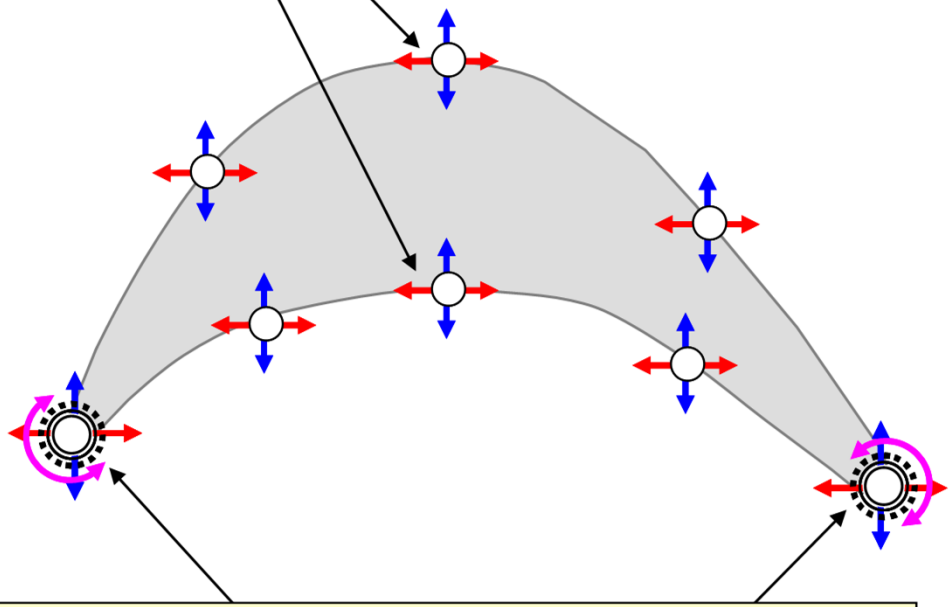


設計変数 design parameters



Blade Surface Control Points

- Radial Position
- Axial Position



Leading / Trailing Edge Shape

- Curvature Radius
- Rotation
- Radial Position
- Axial Position (Except Hub Surface)

Total Design Variables : 58

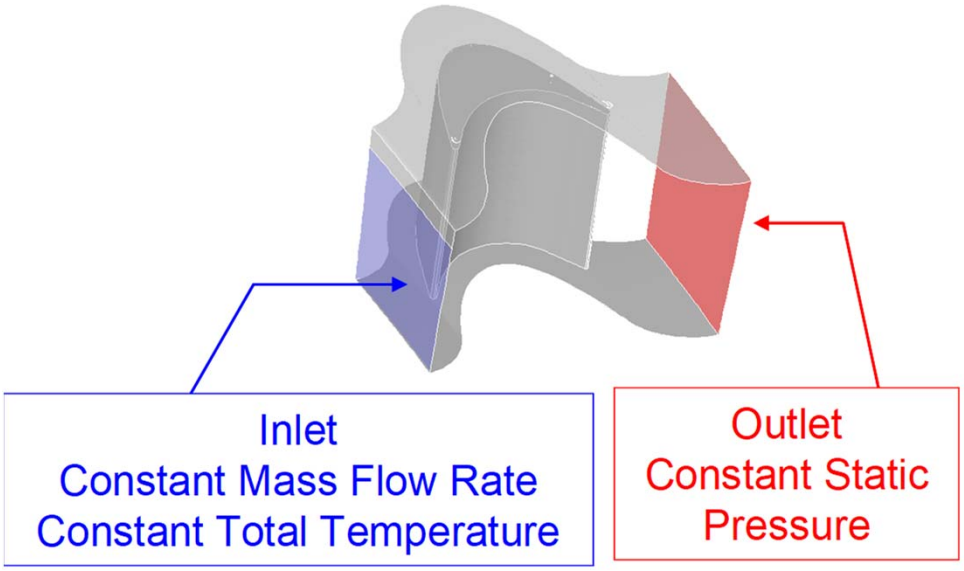
最適化条件 design condition

回転数(rotation speed) : 14,000 rpm
流体 (fluid): 空気 (air)

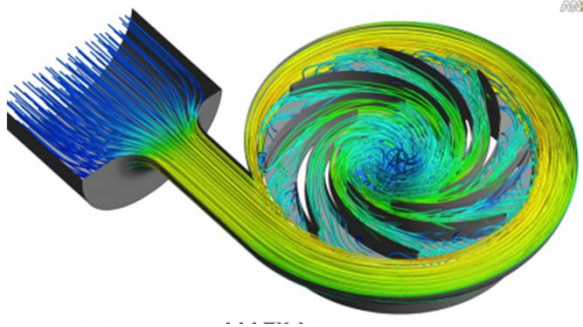
計算手法 Computation methods

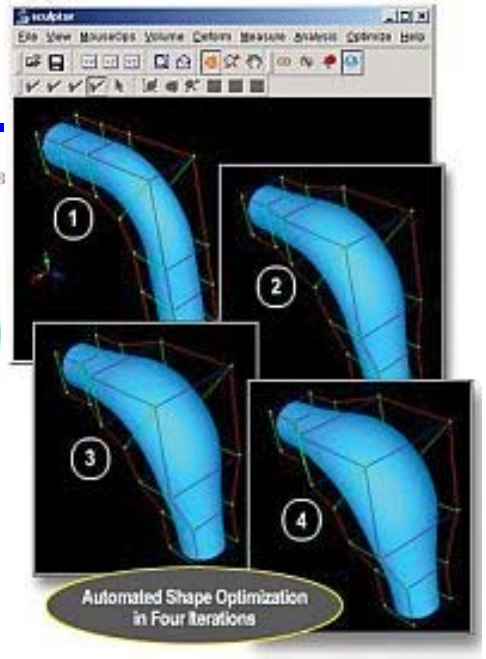
Flow computation : FLUENT 6.3.29
Pressure-Velocity Coupling Method:
Compressible/Incompressible SIMPLE algorithm
Advection Scheme: Second-order Upwing
Turbulence Model: Realizable k-e MODEL+Non-equilibrium wall function

Optimization: Multiobjective evolutionary algorithm
16 Populations x 100 Generations = 1600 CFD runs.



Optimization flowchart





CFD grid generation of nominal design

Grid deformation (SCULPTOR)

CFD computation (FLUENT)

Computation of objective function and constraint values

Converged?

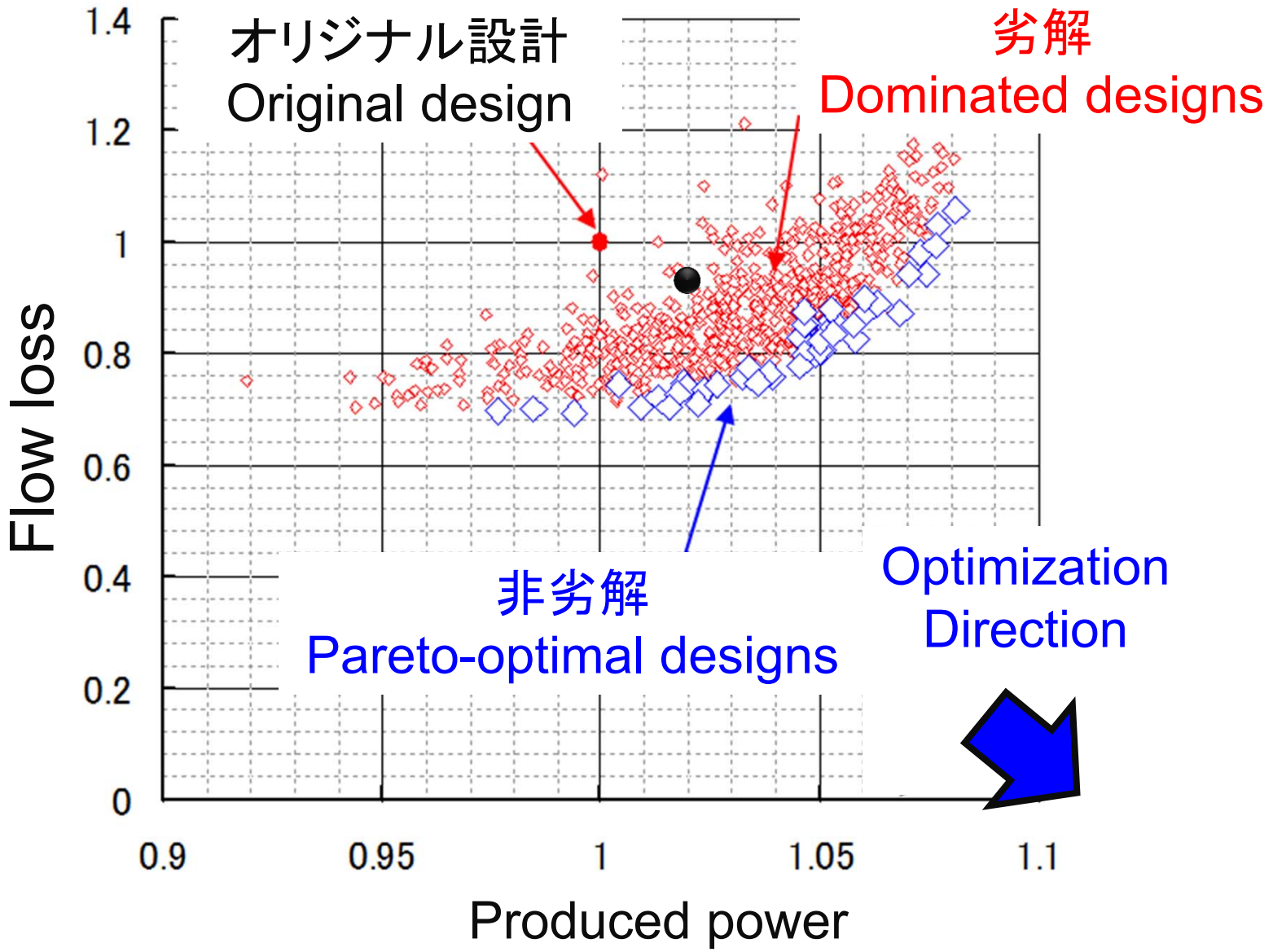
NO

YES

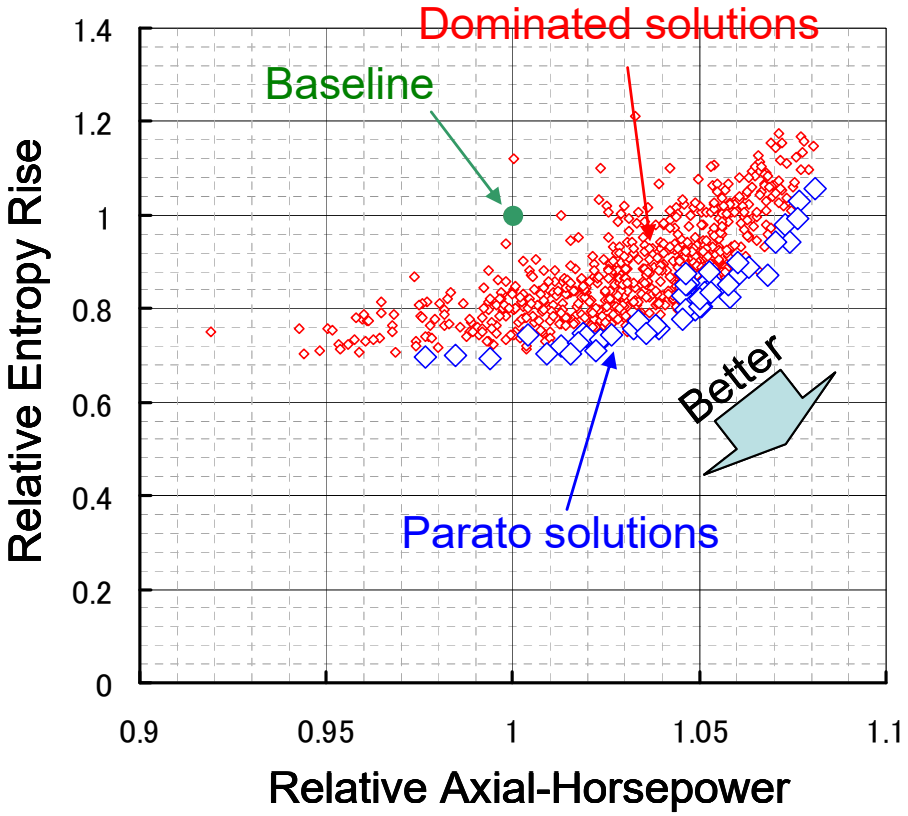
multiobjective evolutionary algorithm

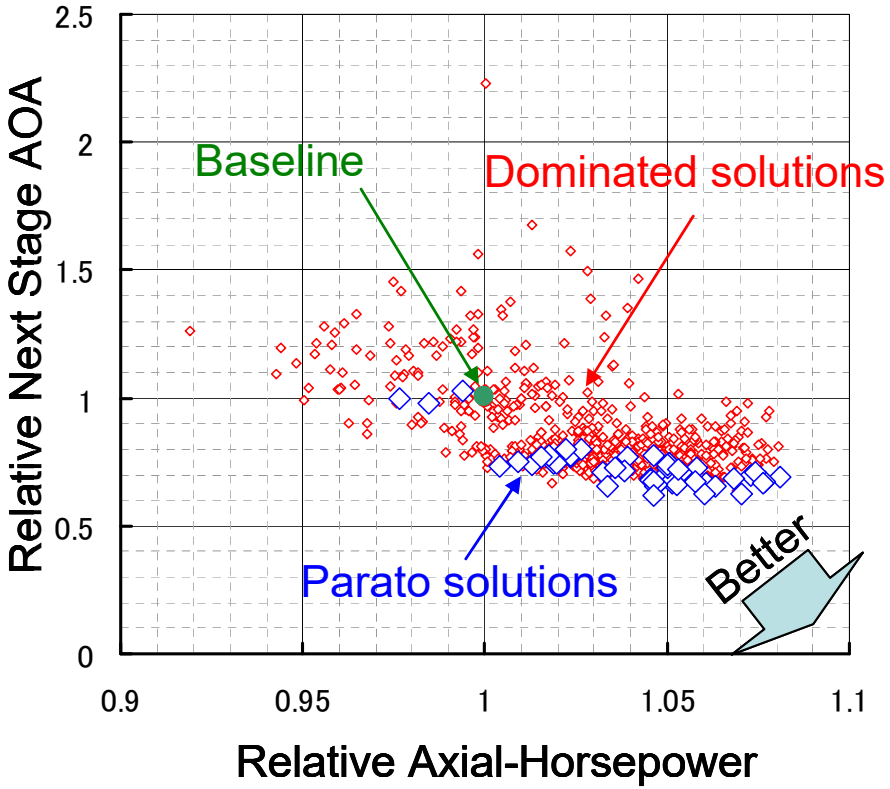
end

Performance of Pareto-optimal designs



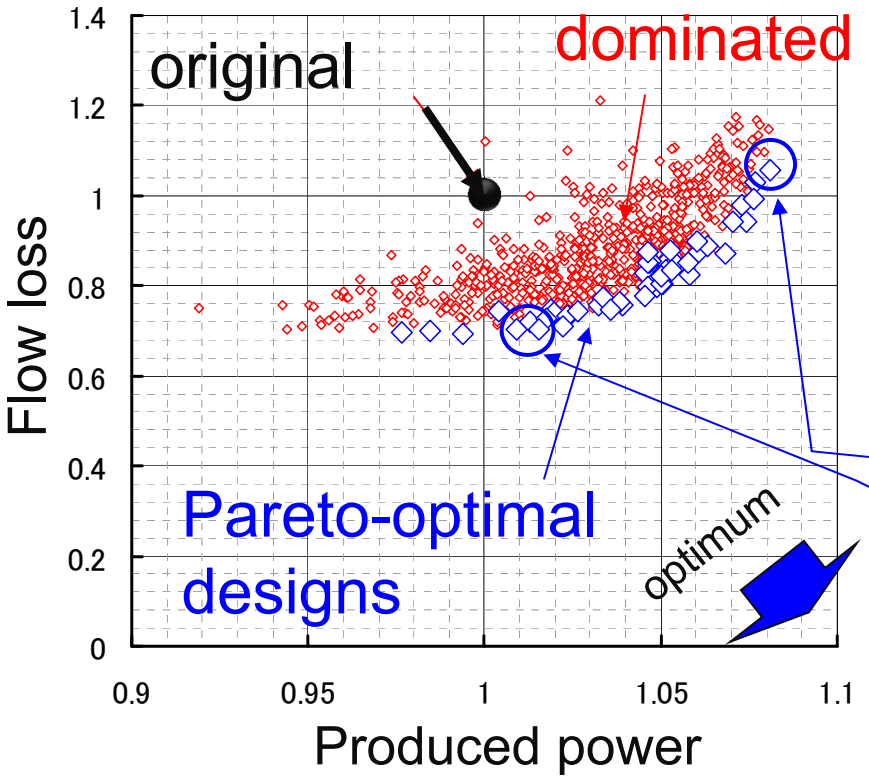
Performance of Pareto-optimal designs

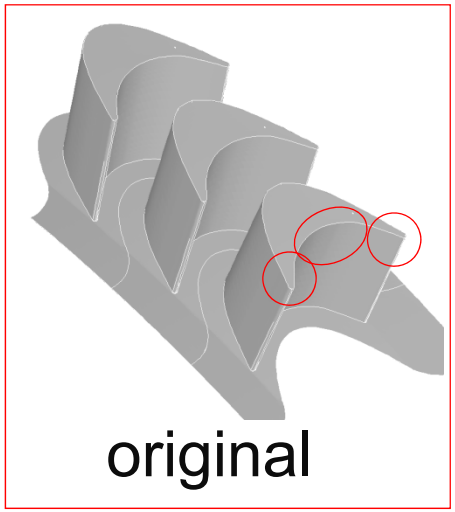


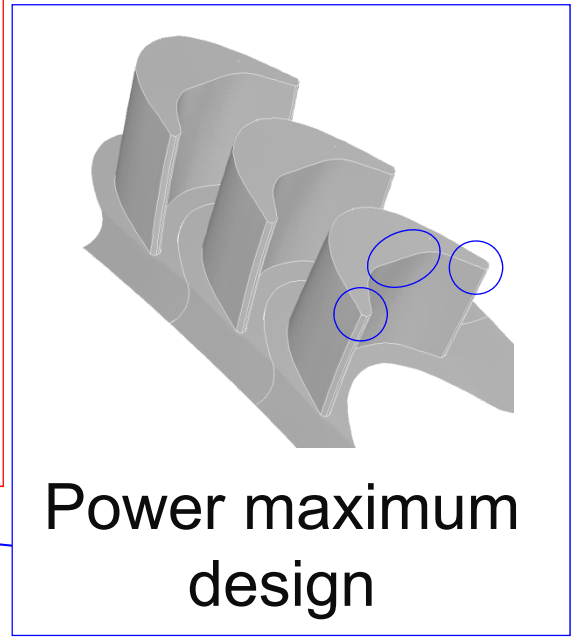


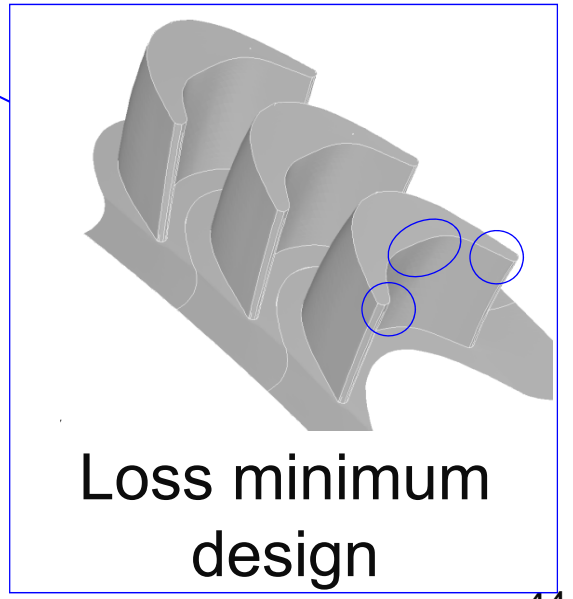
- Designs better than the original design in all objectives were found
- Tradeoff information among objectives was revealed

Analysis of optimized shapes





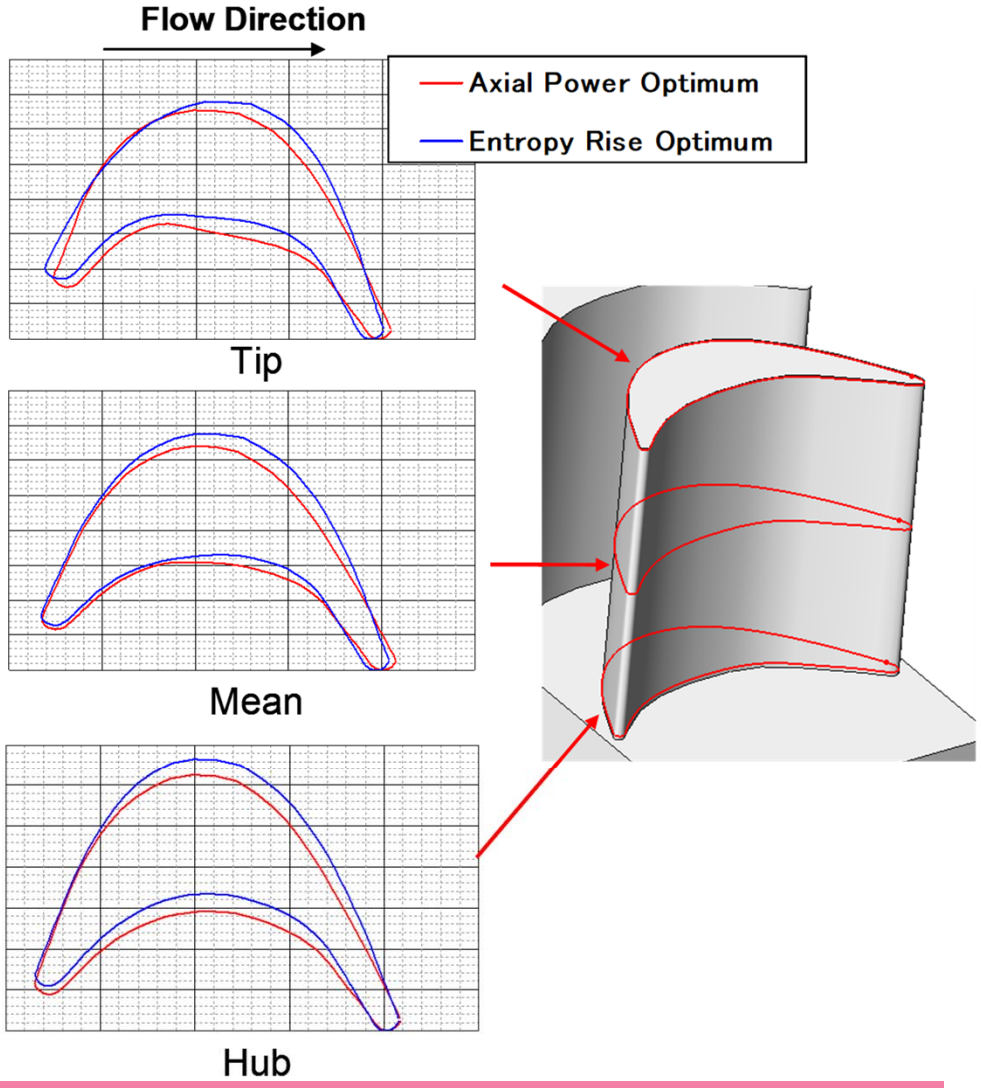


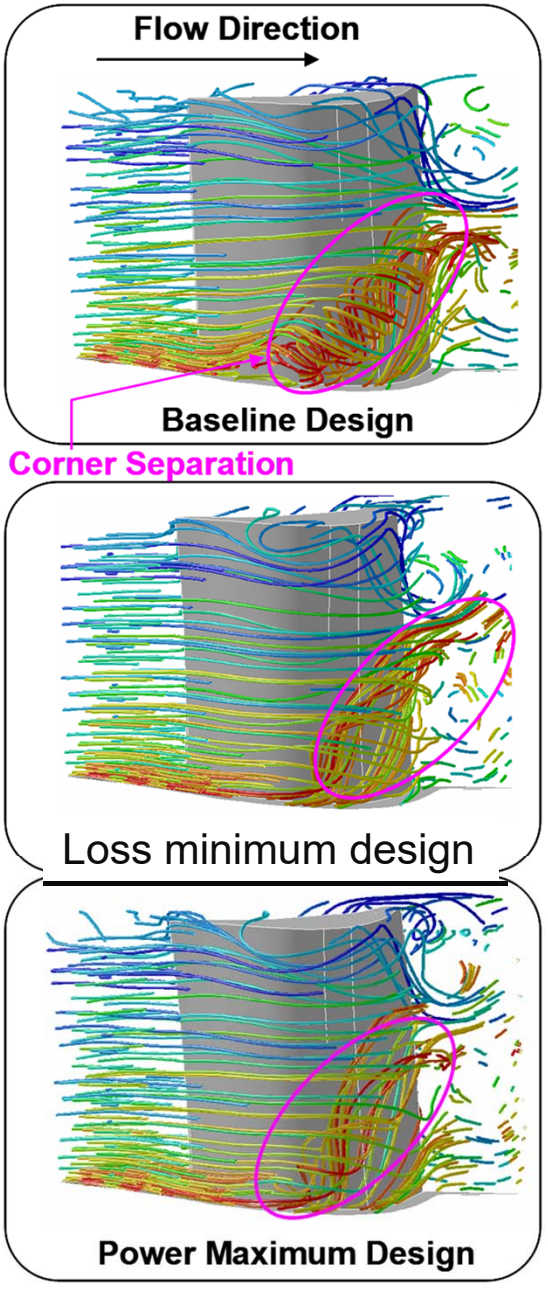


New findings

- leading/trailing edge should be blunt
- Mid-chord should be flat on pressure side

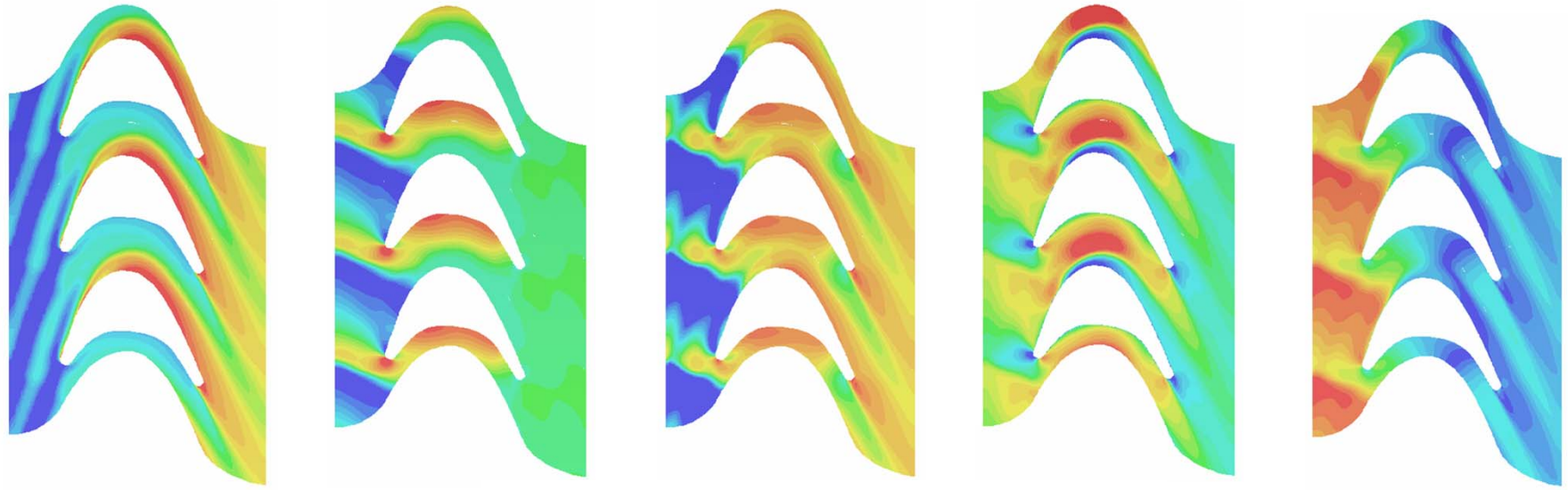
Analysis of flow field





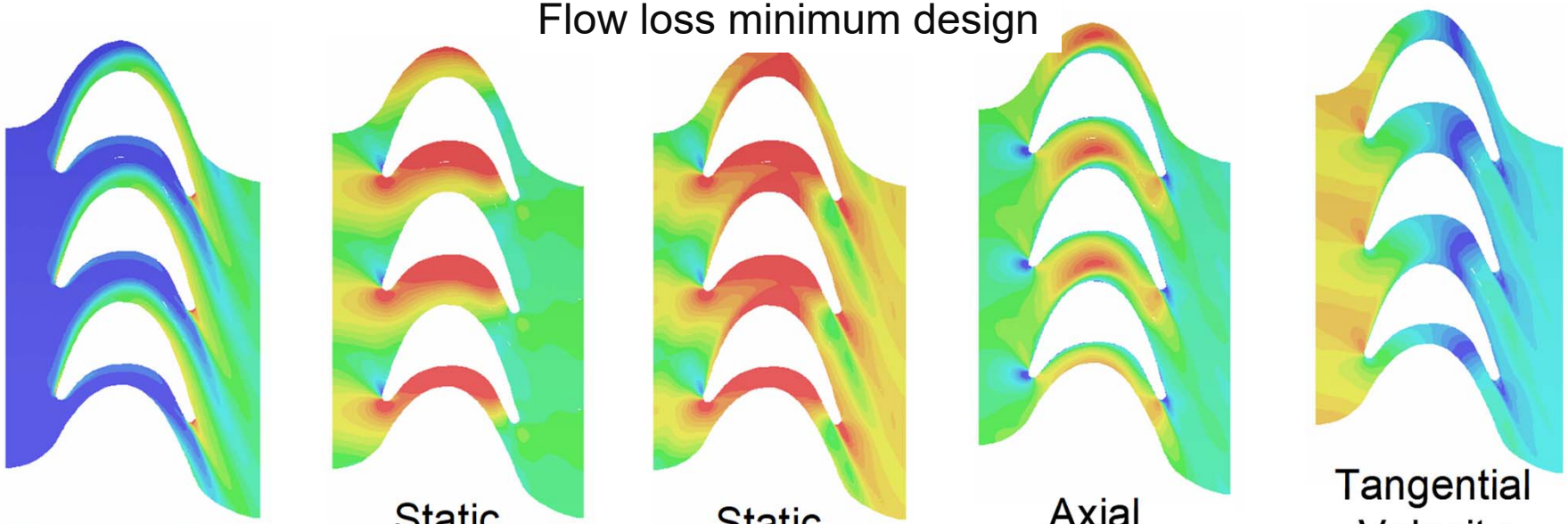
Suppression of corner separation is important for flow loss minimization

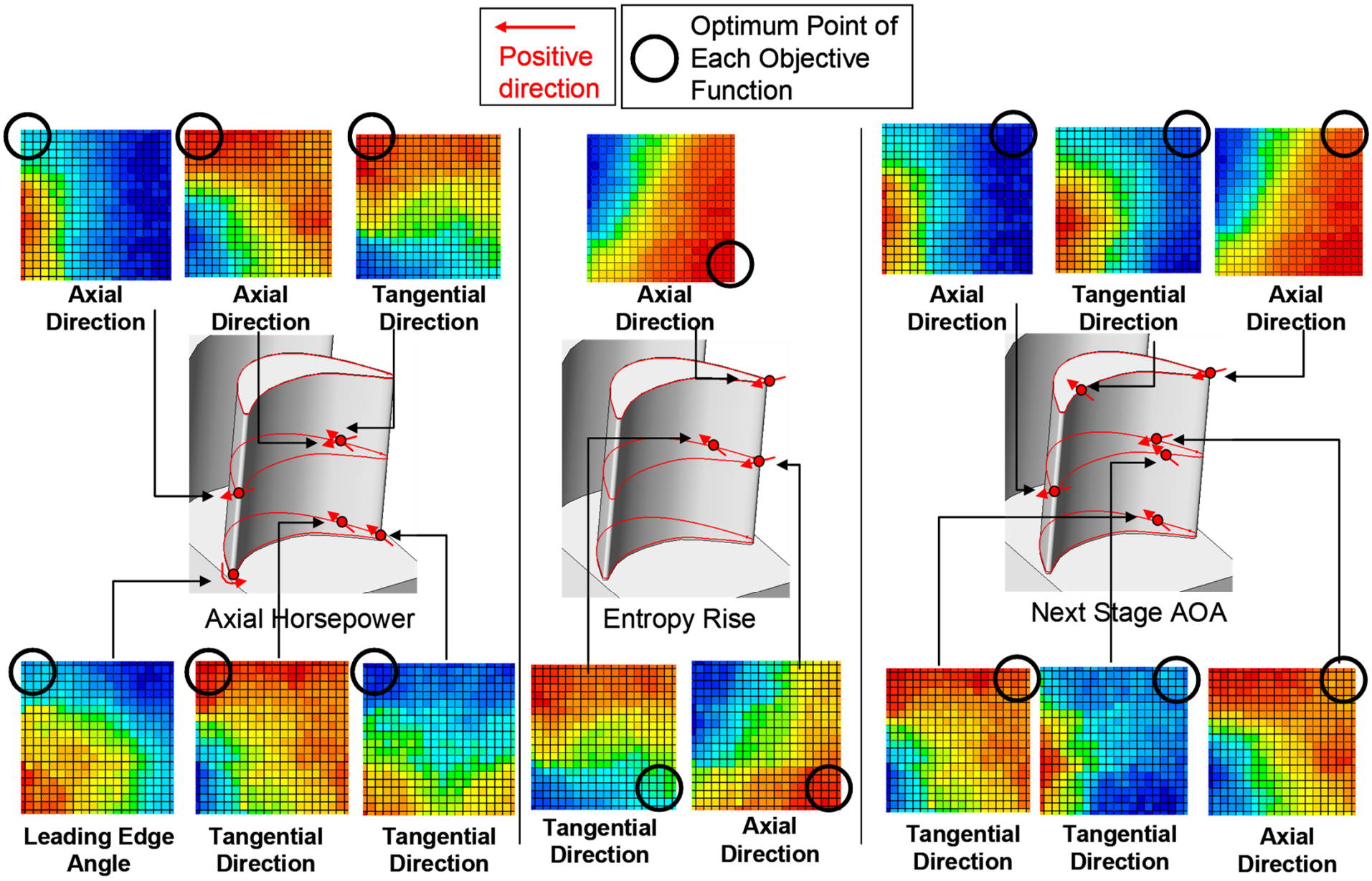
Produced power maximum design has larger tangential velocity magnitude



Produced power maximum design

Flow loss minimum design




Positive direction
Optimum Point of Each Objective Function
Axial Direction
Axial Direction
Tangential Direction
Axial Direction
Axial Horsepower
Entropy Rise
Axial Direction
Tangential Direction
Axial Direction
Next Stage AOA
Leading Edge Angle
Tangential Direction
Tangential Direction
Tangential Direction
Axial Direction
Tangential Direction
Tangential Direction
Axial Direction

Summary

Pump impeller design

4.4% reduction in required power compared with original design maintaining required pressure rise

Turbine blade design

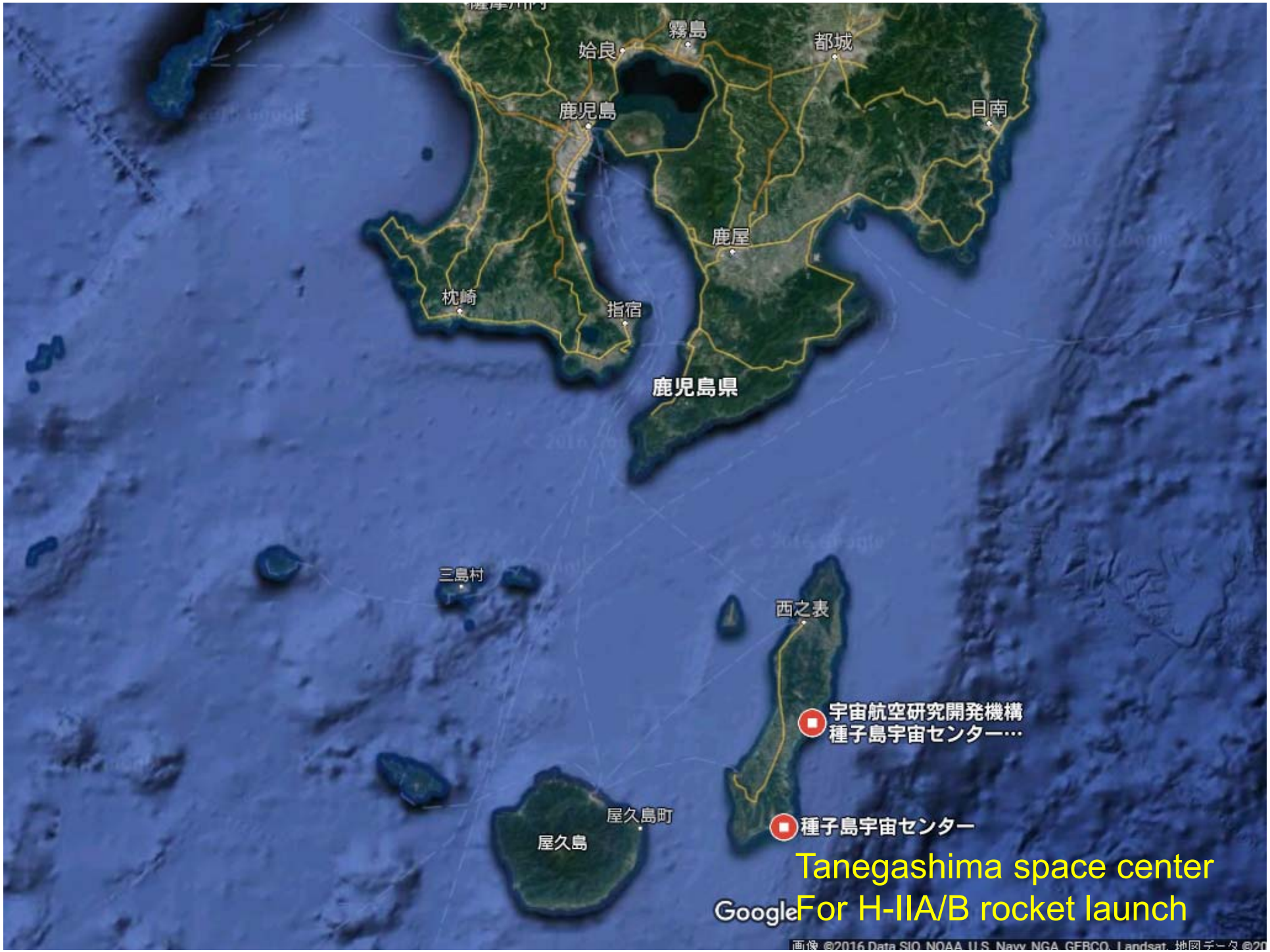
8% increase in produced power compared with original design

Designs that have more than 10% improvement in efficiency are found

New knowledge for efficient turbopump design

Aeroacoustic design optimization of rocket launch site

ロケット射点形状の音響環境最適化

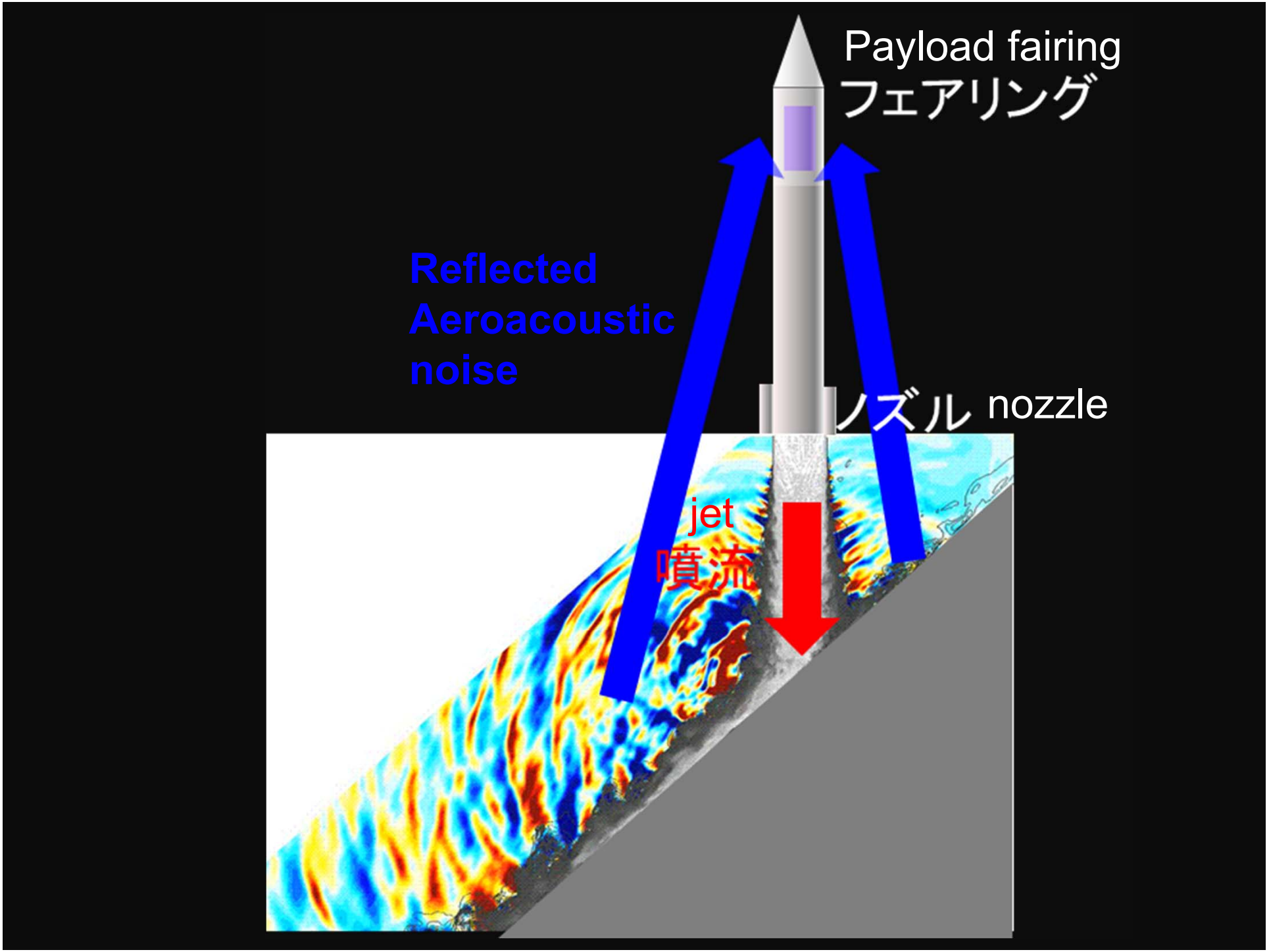

姶良
霧島
都城
鹿児島
日南
鹿屋
枕崎
指宿
鹿児島県
三島村
西之表
宇宙航空研究開発機構
種子島宇宙センター…
屋久島町
屋久島
種子島宇宙センター
Tanegashima space center
For H-IIA/B rocket launch
Google
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Uchinoura space center
for epsilon rocket launch




EPSILON


Payload fairing
フェアリング
Reflected
Aeroacoustic
noise
ノズル nozzle
jet
噴流

M-Vロケットの打ち上げ
＠内之浦宇宙空間観測所

M-V rocket launch
@Uchinoura Space Center






Flame deflector



Flame deflector







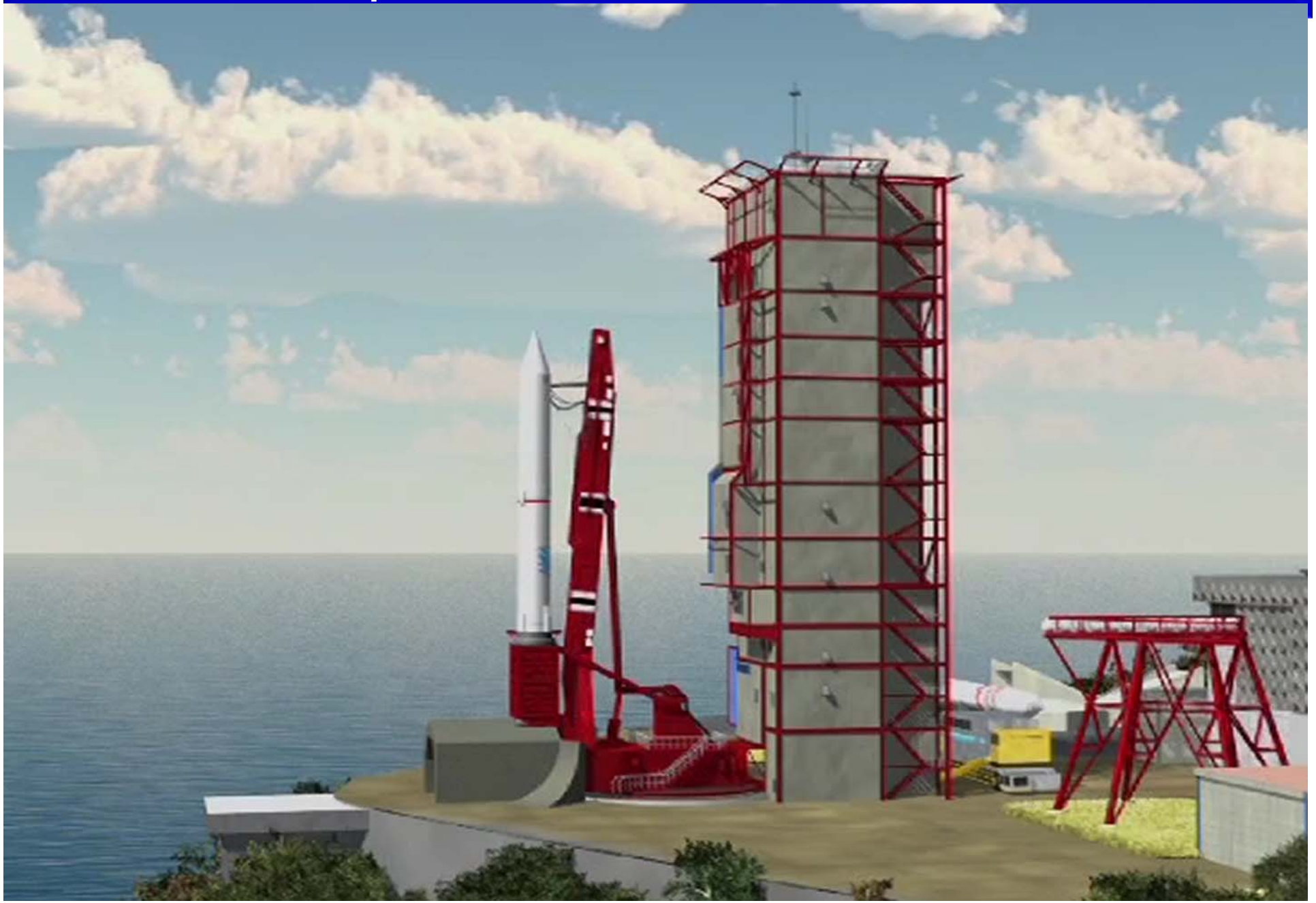
イプシロンロケットの打ち上げ
＠内之浦宇宙空間観測所

Epsilon rocket launch
@Uchinoura Space Center

Epsilon rocket launch site



Epsilon rocket launch site






M-V-7



Epsilon rocket launch site



Epsilon rocket launch site



なにがかわったでしょうか？

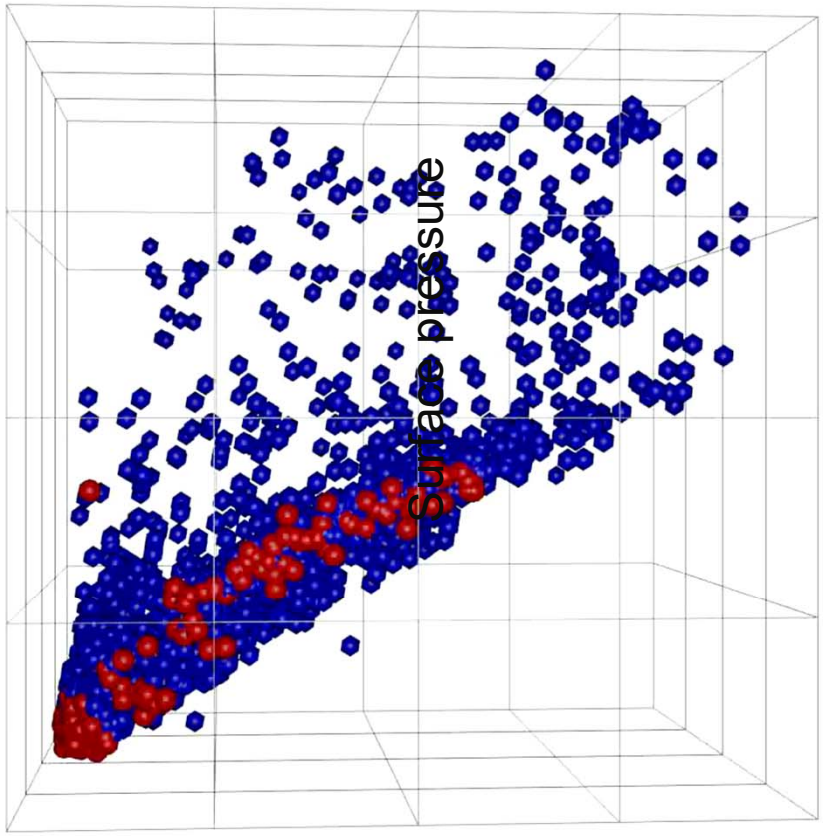


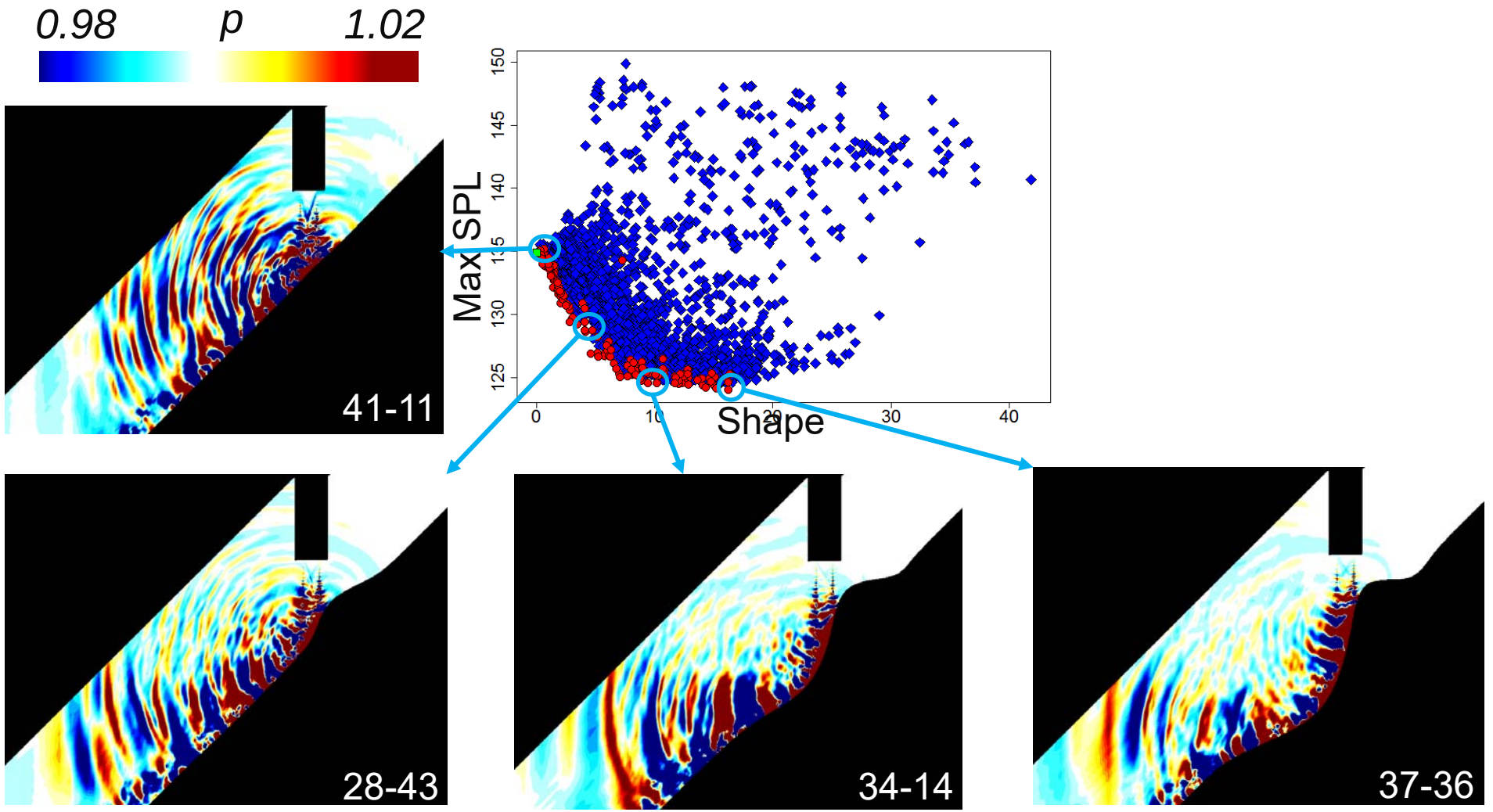


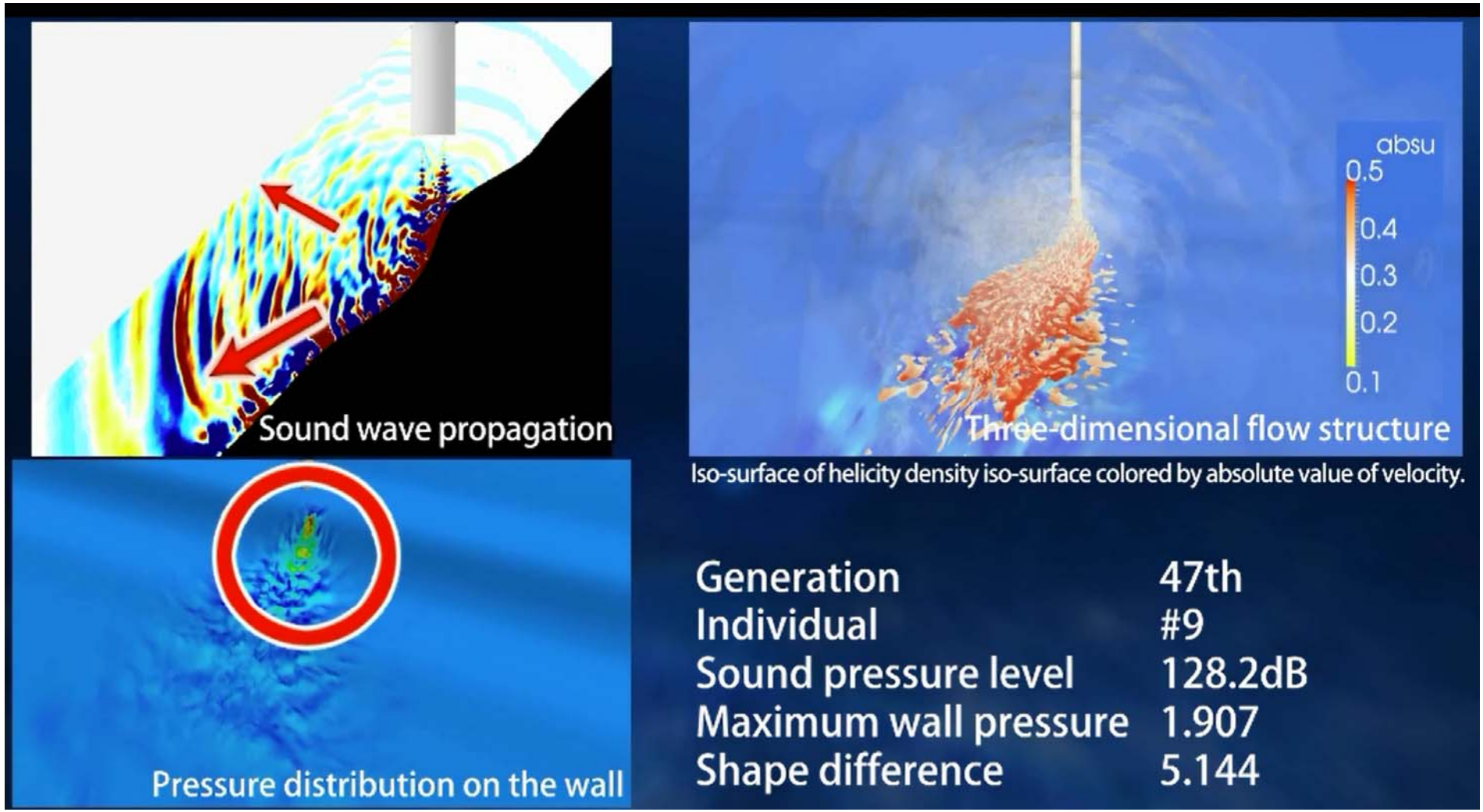
movie



Tradeoff among objectives




0.98
p
1.02
41-11
Max SPL
150
145
140
135
130
125
0
10
20
30
40
Shape
28-43
34-14
37-36



Making a small bump at upstream of the impinging point reduce reflected aeroacoustic noise significantly

Movie is available on YouTube

日本語版
www.youtube.com/watch?v=Y2bo5CbXQ3w
ロケット射点の空力音響多目的設計探査の研究
「多目的設計探査」で検索

English version
http://www.youtube.com/watch?v=y4mx4ucd_-k
Search for “multiobjective design exploration”